The Cell

Plan for the day

- Cell:
  - Prokaryotic vs. Eukaryotic
  - Plant vs. Animal
  - Organelles
  - Cytoskeleton
  - Cell membrane
  - Cell adhesion
  - Cell-cell communication

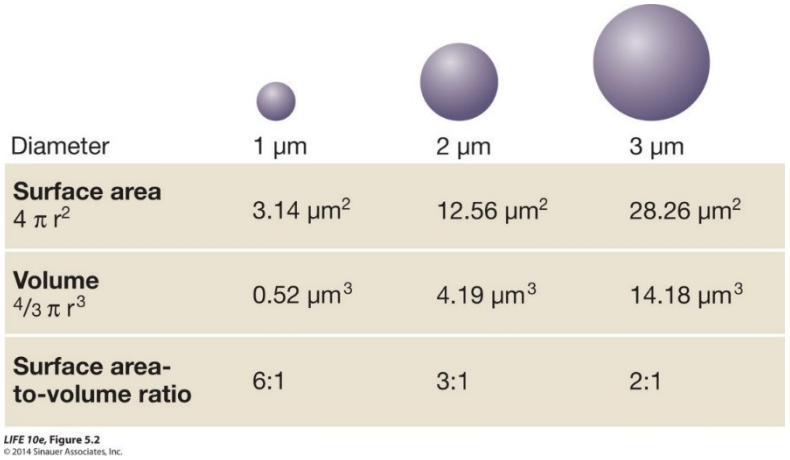
Cell theory

- Cell theory was the first unifying theory of biology:
  - Cells are the fundamental units of life.
  - All organisms are composed of cells.
  - All cells come from preexisting cells.
- Implications of the cell theory:
  - Functions of all cells are similar.
  - Life is continuous.
  - Origin of life was the origin of cells.

Cell size

- Cells are small because a high surface area-to-volume ratio is essential.
- Volume determines the amount of chemical activity in the cell per unit time. Larger cells have more chemical activity.
- Cell surface area limits the amount of resources and waste products that can cross the cell boundary per unit time.

Why Cells Are Small?



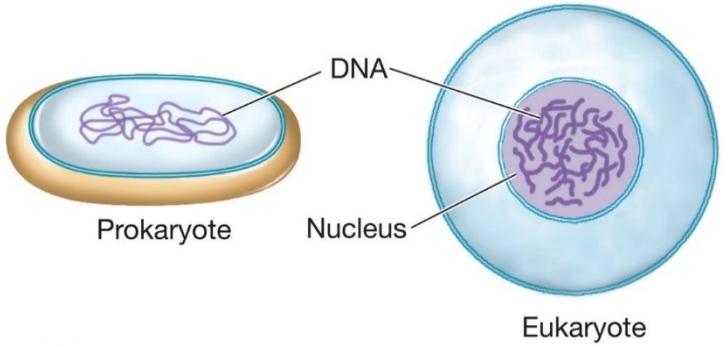
Diameter	1 µm	2 µm	3 µm
Surface area $4 \pi r^2$	3.14 μm^2	12.56 μm^2	28.26 μm^2
Volume $4/3 \pi r^3$	0.52 μm^3	4.19 μm^3	14.18 μm^3
Surface area-to-volume ratio	6:1	3:1	2:1

LIFE 10e, Figure 5.2
© 2014 Sinauer Associates, Inc.

1. Area to volume ratio needs to be kept high.
2. Diffusion rate of molecules is a limiting factor.
3. Local concentrations should be kept high for most enzymatic reactions.

Prokaryotes vs. Eukaryotes

- Two types of cells: **Prokaryotic** and **eukaryotic**.
- Bacteria and Archaea are **prokaryotes**. They have no membrane-enclosed internal compartments.



LIFE 10e, In-Text Art, Ch. 5, p. 81
© 2014 Sinauer Associates, Inc.

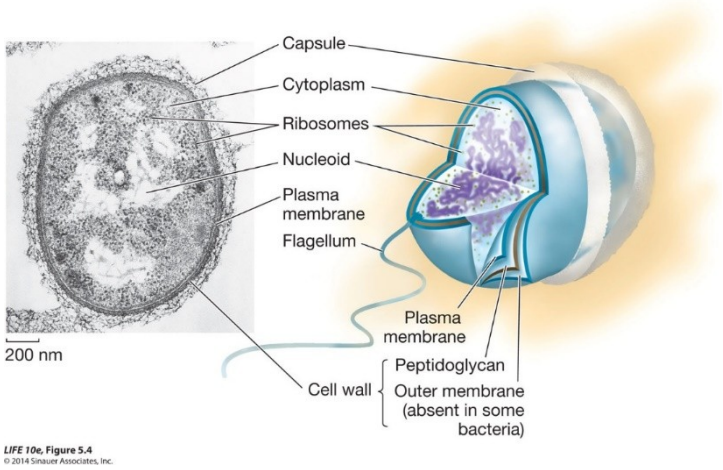
Prokaryotes vs. Eukaryotes

- Prokaryotic cells are very small.
- Individuals are single cells but often form chains or clusters.
- Prokaryotes are very successful; and there is a huge diversity of species in the Bacteria and Archaea domains.

Prokaryotes vs. Eukaryotes

- **Eukaryotic** cells with membrane-enclosed compartments called **organelles**.
- The DNA is in a compartment called the **nucleus**. Specific chemical reactions occur in other organelles.
- This “division of labor” was important in the evolution of complex organisms.

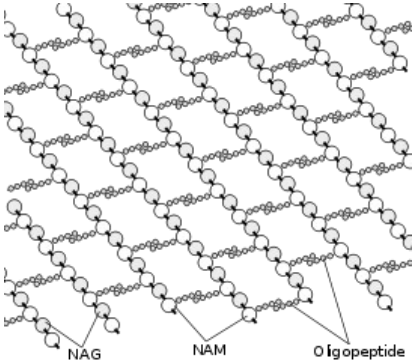
A Prokaryotic Cell



LIFE 10e, Figure 5.4
© 2014 Sinauer Associates, Inc.

Prokaryotic cell

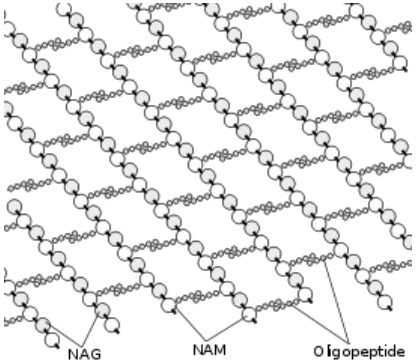
- Most prokaryotes have a rigid **cell wall** outside the plasma membrane.
- Bacterial cell walls contain peptidoglycan (peptide+sugar).
- Bacterial cell wall forms a mesh layer over their plasma membrane.



Bacterial cell wall

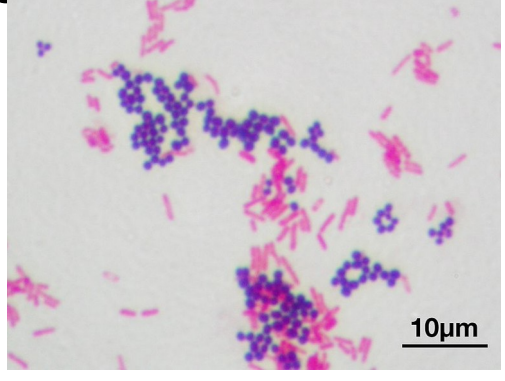
Penicillin interferes with bacterial cell wall formation

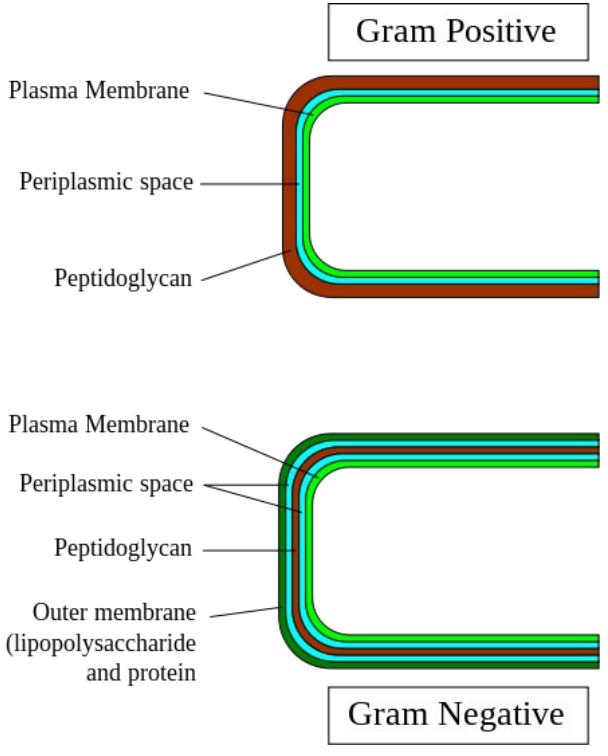
- Penicillin blocks the enzyme that links the mesh together.
- If no cell wall→ Bacteria burst due to osmotic pressure
- https://www.youtube.com/watch?v=UjLmf-cVcMw



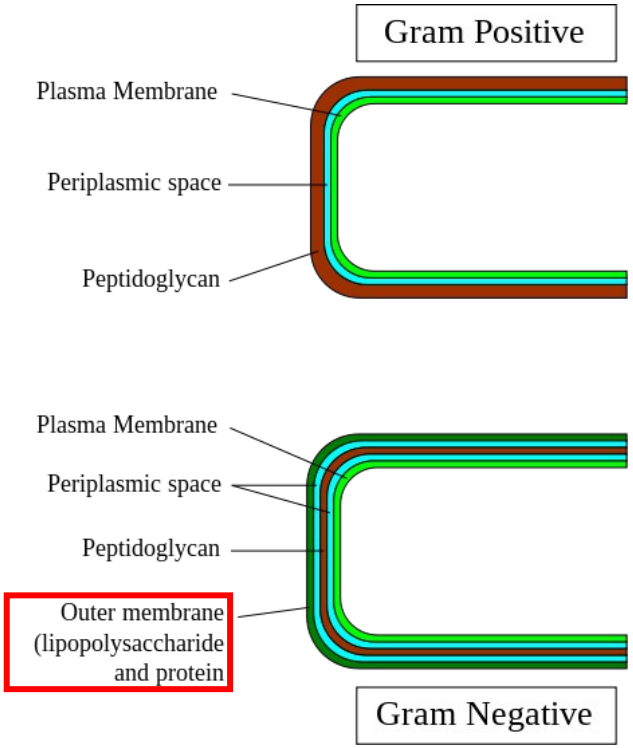
Categorizing bacteria

- Some bacteria have an additional **outer membrane**.
- Gram staining of bacteria is used to categorize bacteria fast and cheaply (Gram+ and Gram-)
- Thickened peptideoglycan layer retains Gram stain better→ Gram+
- <u>Gram- bacteria have an outer membrane</u> but thin cell wall so they stain lighter

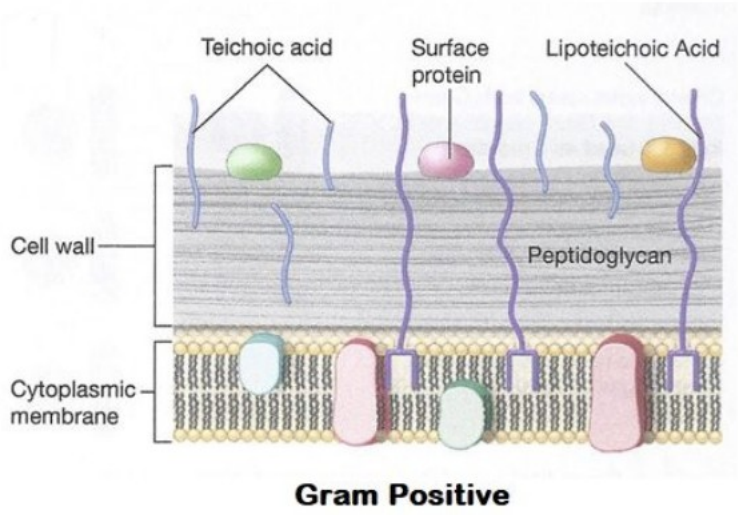




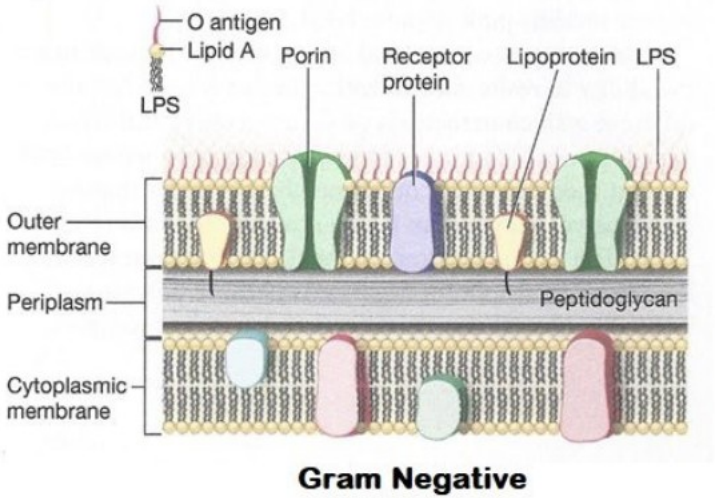
Does penicillin work on gram positive or gram negative bacteria?



Does penicillin work on gram positive or gram negative bacteria?



Gram Positive



Gram Negative

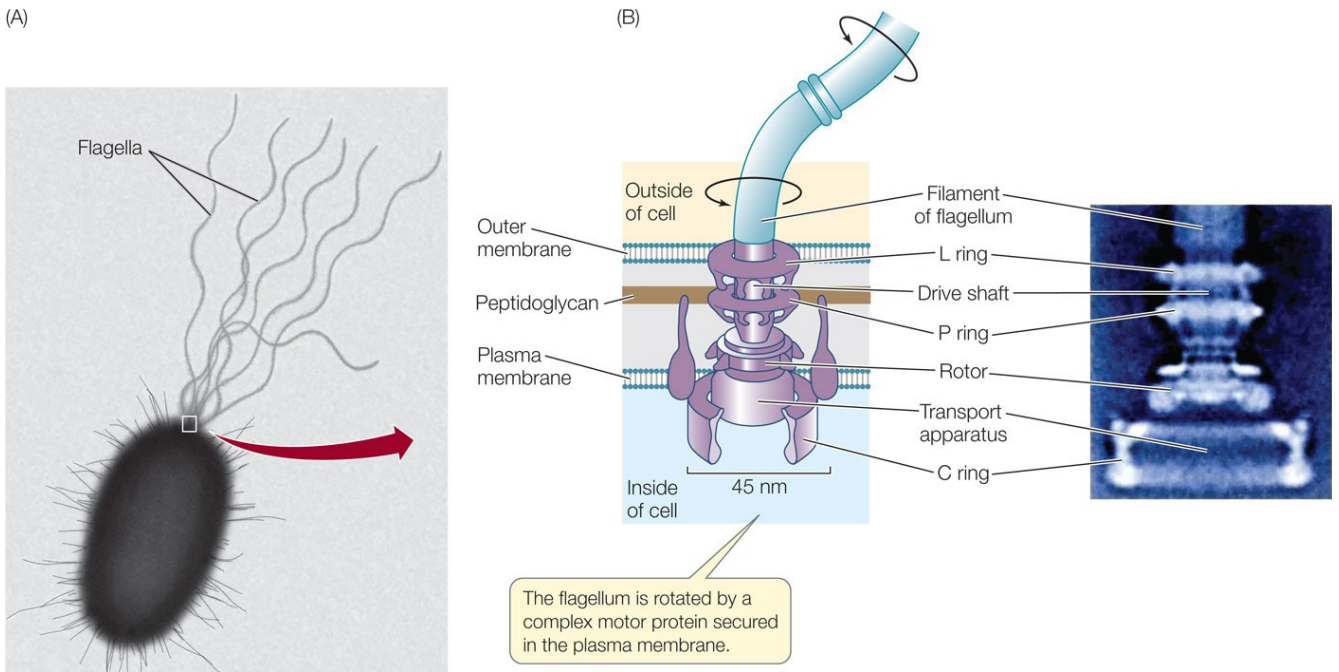
Prokaryotic Cell

- Photosynthetic bacteria have an **internal membrane system** that contains molecules necessary for photosynthesis.
- Others have internal membrane folds that are attached to the plasma membrane; they may function in cell division or in energy-releasing reactions.

Prokaryotic Cell

- Some prokaryotes swim by means of **flagella**, made of the protein flagellin.
- Some bacteria have **pili**—hairlike structures projecting from the surface. They help bacteria adhere to other cells.
- Fimbriae are shorter than pili; they help cells adhere to surfaces such as animal cells.

Prokaryotic Flagella



***LIFE 10e*, Figure 5.5**

© 2014 Sinauer Associates, Inc.

Eukaryotic Cells

- Eukaryotic cells are up to ten times larger than prokaryotes.
- Eukaryotic cells have membrane-enclosed compartments called **organelles**.
- Each organelle has a specific role in cell functioning.

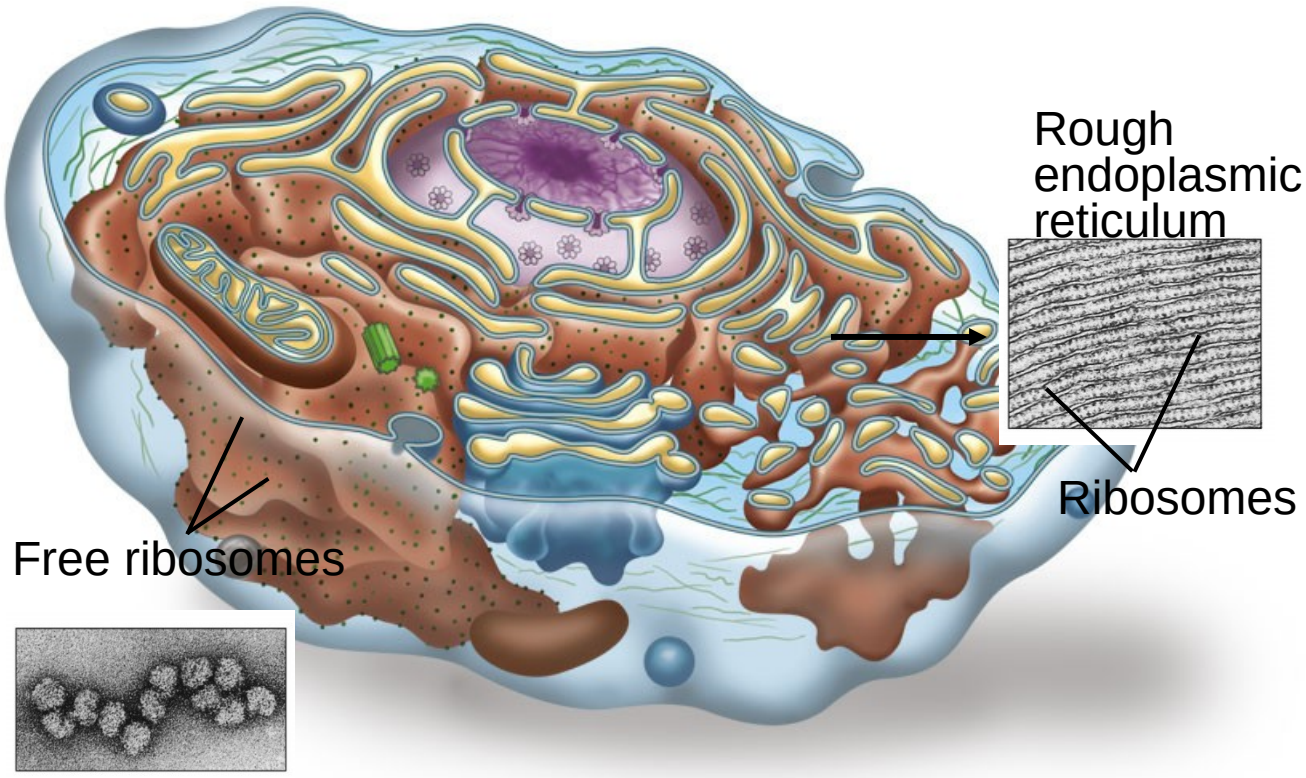
Eukaryotic Cells

- Compartmentalization allowed eukaryotic cells to specialize and form the tissues and organs of multicellular organisms.
- To determine the functions of organelles, they were first studied using light microscopy and then electron microscopy.
- Cell fractionation separates organelles by size or density for study by chemical methods.

Ribosomes: Site for protein synthesis

- Ribosomes: sites of protein synthesis.
- Occur in both prokaryotic and eukaryotic cells and have similar structure.
- Ribosomes consist of ribosomal RNA (rRNA) and more than 50 different protein molecules.
- In eukaryotes, ribosomes are free in the cytoplasm, attached to the endoplasmic reticulum, or inside mitochondria and chloroplasts.
- In prokaryotic cells, ribosomes float freely in the cytoplasm.

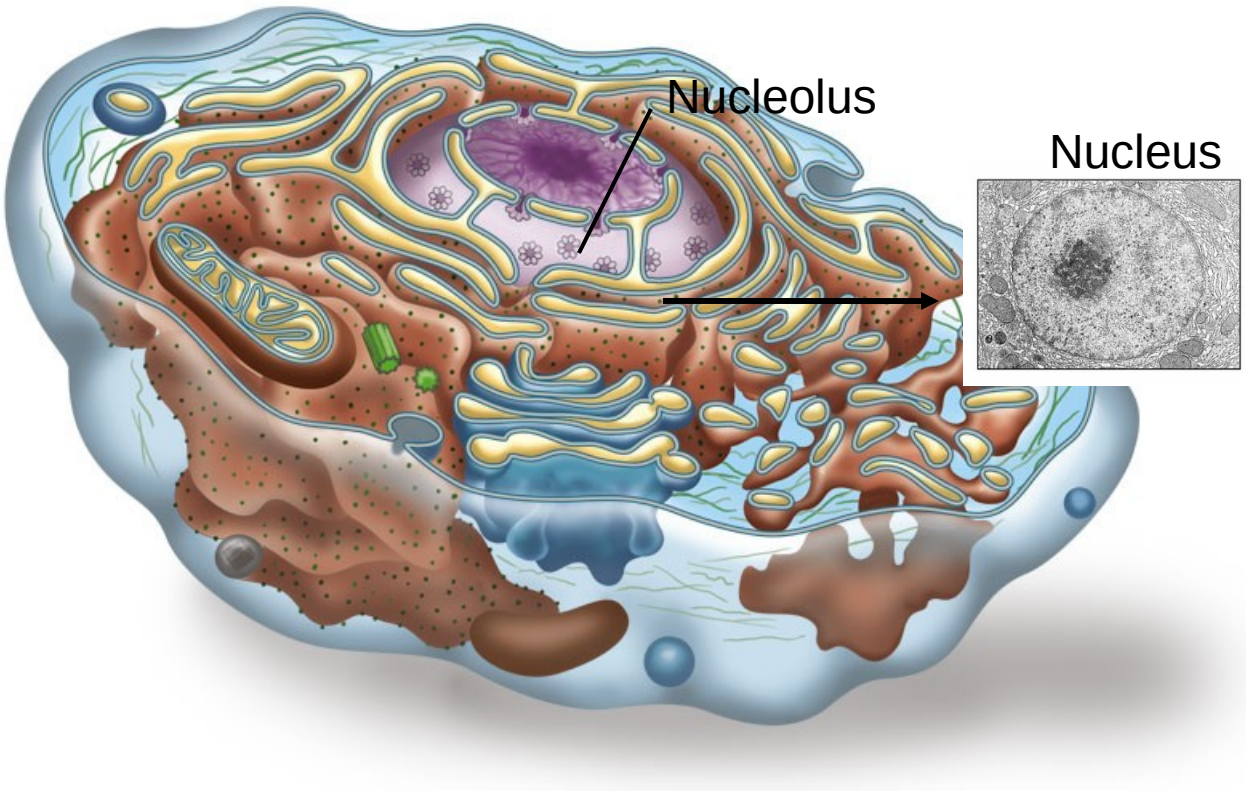
The rough endoplasmic reticulum is the site of protein synthesis.



Nucleus: The control room

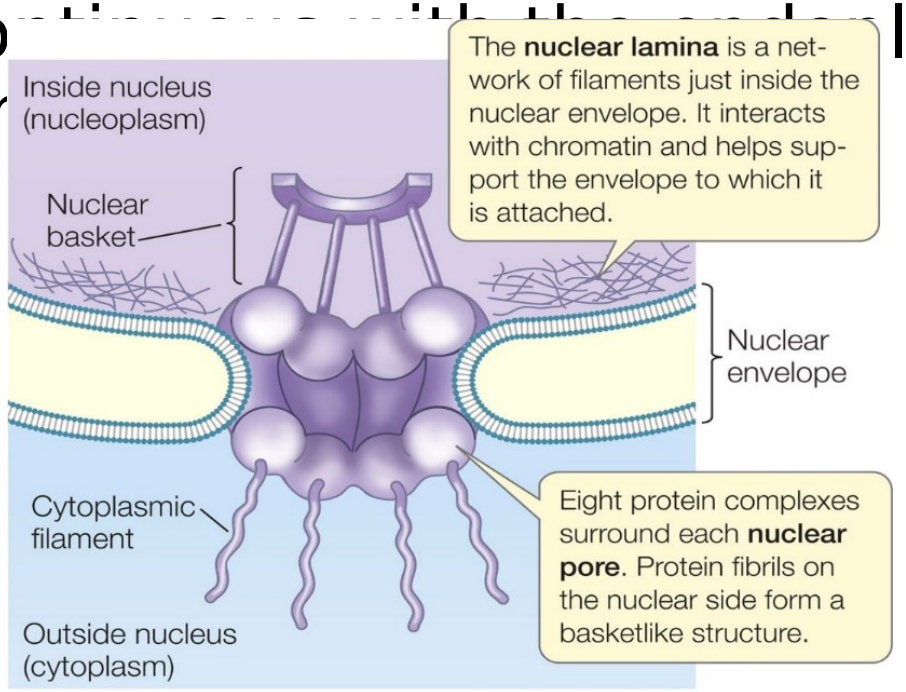
- The nucleus is usually the largest organelle.
- Contains the DNA
- Site of DNA replication
- Site where gene transcription is turned on or off
- Assembly of ribosomes begins in a region called the **nucleolus**

The nucleus is the site of most cellular DNA, which, with associated proteins, comprises chromatin.



Nucleus

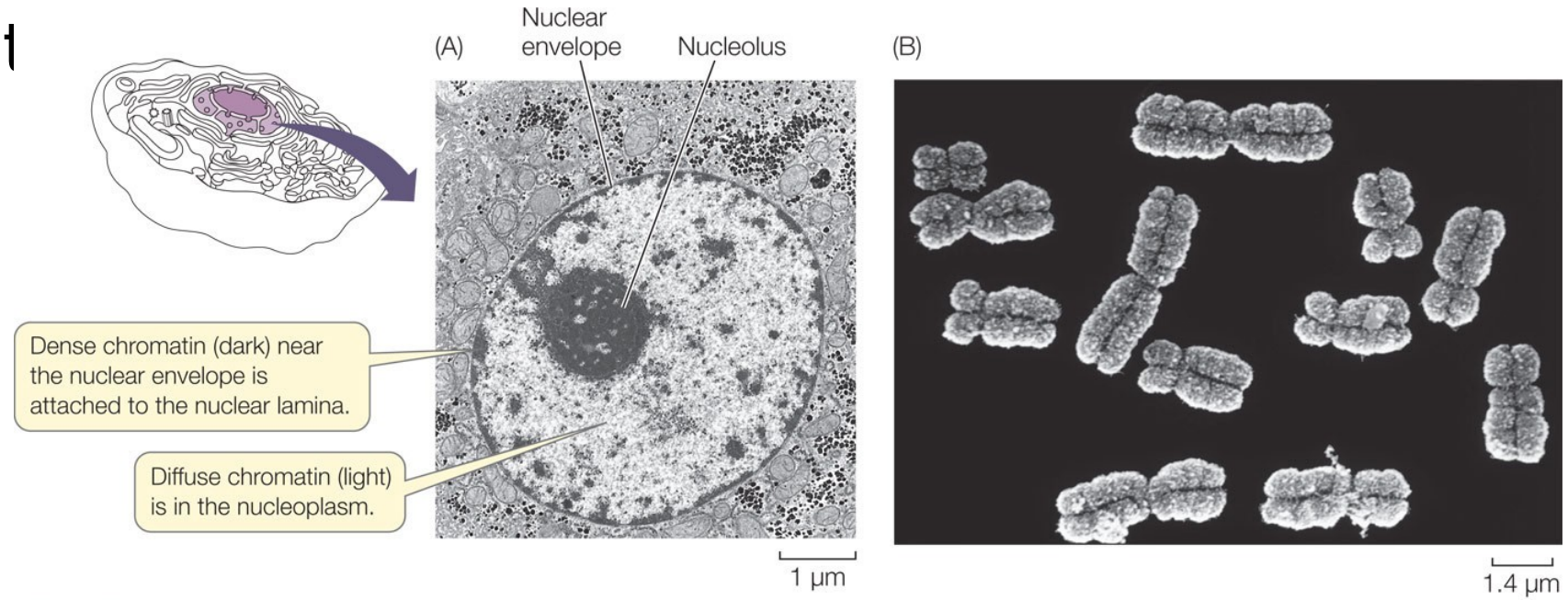
- The nucleus is surrounded by the **nuclear envelope**. Many pores control the movement of molecules across the envelope.
- The outer membrane of the nuclear envelope folds outward into the cytoplasm and is co[illegible]lasmic reticulum



LIFE 10e, In-Text Art, Ch. 5, p. 85
© 2014 Sinauer Associates, Inc.

Nucleus

- In the nucleus, DNA combines with proteins to form **chromatin** in long, thin threads called **chromosomes**.
- Before cell division, chromatin condenses, and individual chromosomes are visible in t



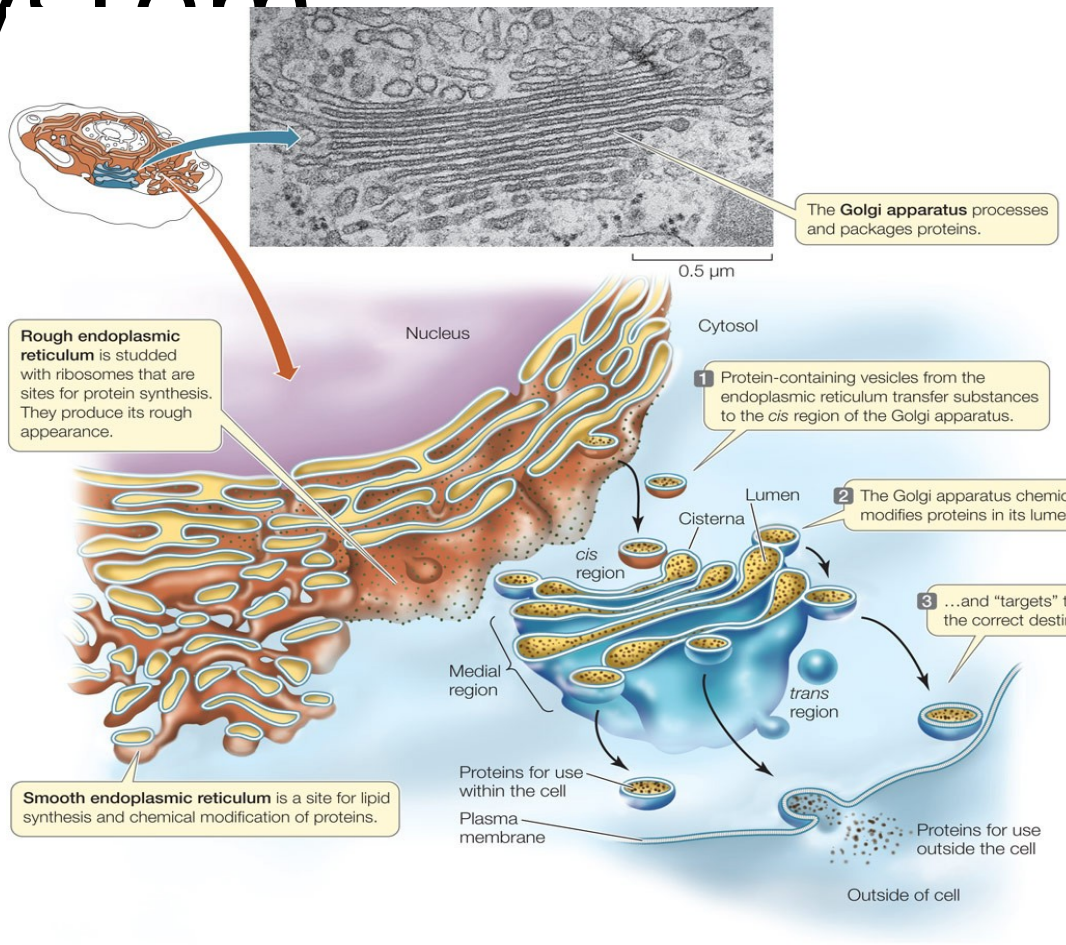
***LIFE 10e*, Figure 5.8**

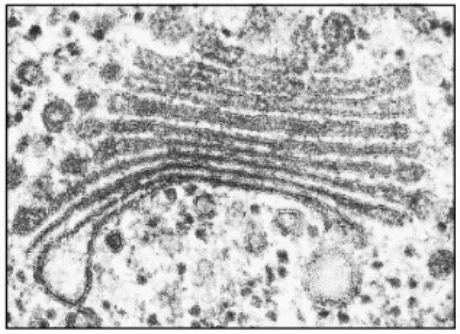
© 2014 Sinauer Associates, Inc.

ER and Golgi

- The **endomembrane system** includes the plasma membrane, nuclear envelope, endoplasmic reticulum, Golgi apparatus, and lysosomes.
- Tiny, membrane-surrounded vesicles shuttle substances between the various components.

The Endomembrane System





***LIFE 10e*, Figure 5.9**

© 2014 Sinauer Associates, Inc.

ER

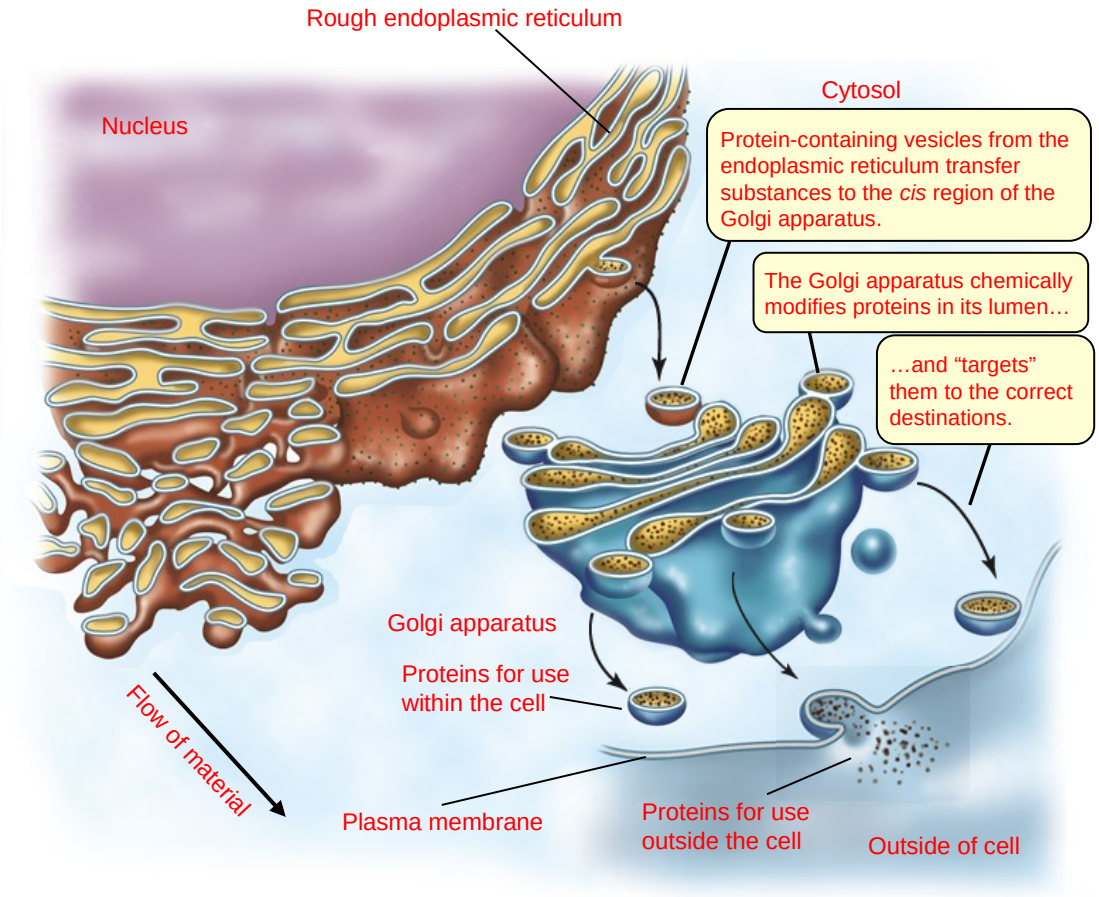
- **Endoplasmic reticulum** (**ER**): network of interconnected membranes in the cytoplasm; has large surface area.
- **Rough endoplasmic reticulum** (**RER**): ribosomes are attached.
- Newly made proteins enter the RER lumen where they are modified, folded, and transported to other regions.

ER

- **Smooth endoplasmic reticulum** (**SER**): more tubular, no ribosomes.
- Chemically modifies small molecules such as drugs and pesticides
- Site of glycogen degradation in animal cells
- Synthesis of lipids and steroids

Golgi

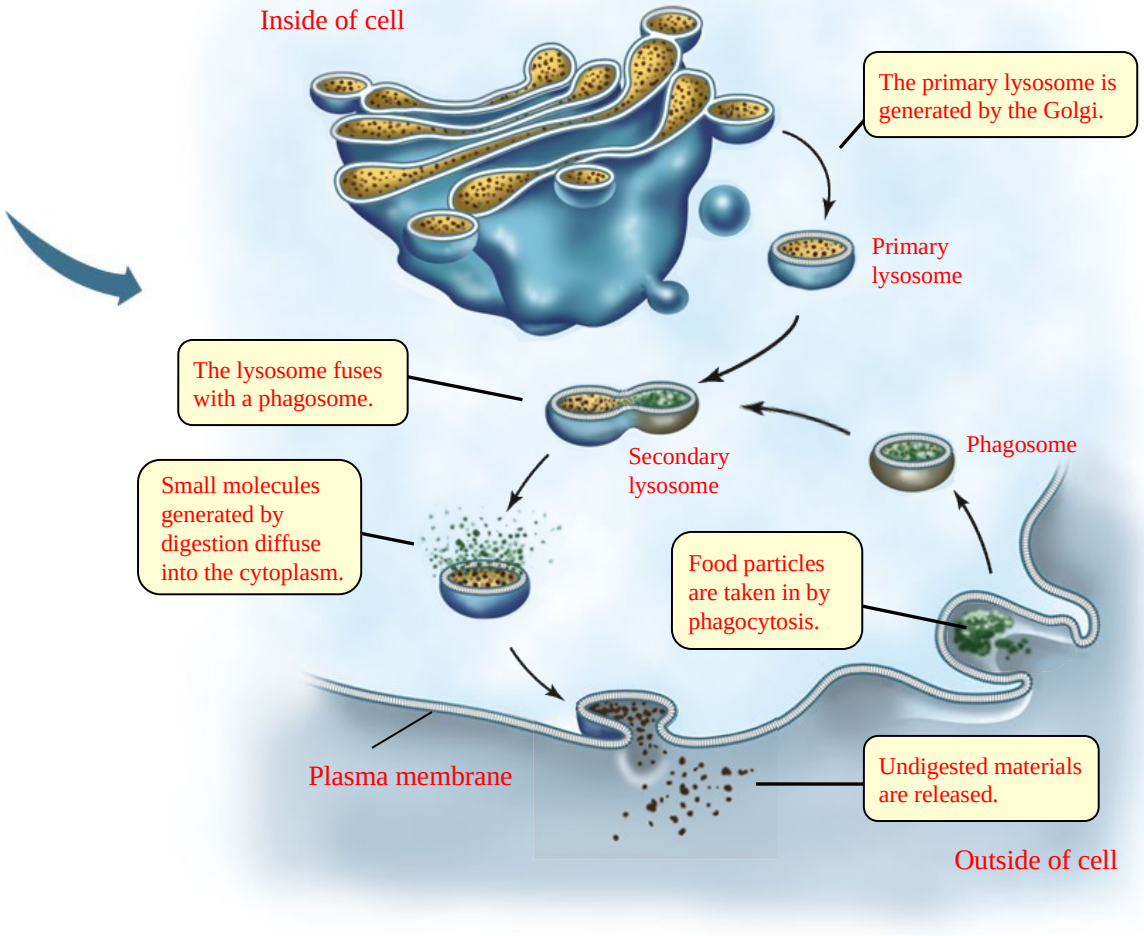
- The **Golgi apparatus** is composed of flattened sacs (cisternae) and small membrane-enclosed vesicles.
- Receives proteins from the RER—can further modify them
- Concentrates, packages, sorts proteins
- In plant cells, polysaccharides for cell walls are synthesized here


Rough endoplasmic reticulum
Nucleus
Cytosol
Protein-containing vesicles from the endoplasmic reticulum transfer substances to the *cis* region of the Golgi apparatus.
The Golgi apparatus chemically modifies proteins in its lumen…
…and "targets" them to the correct destinations.
Golgi apparatus
Proteins for use within the cell
Flow of material
Plasma membrane
Proteins for use outside the cell
Outside of cell

Lysosomes

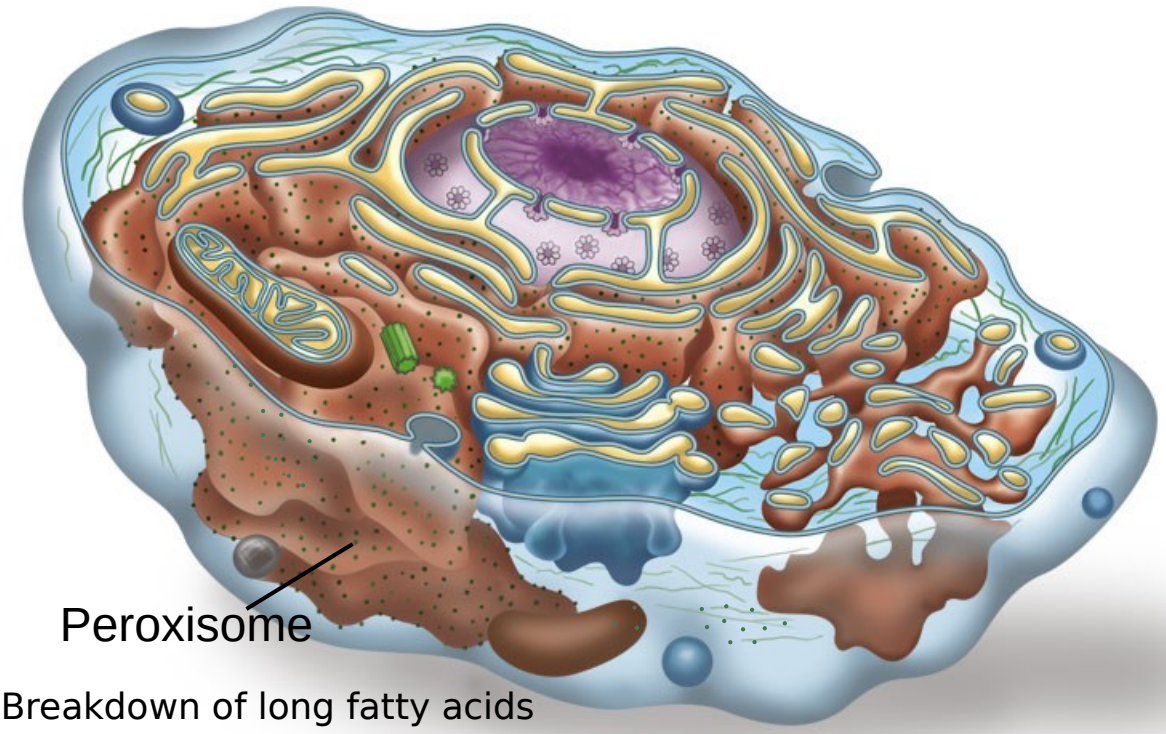
- **Primary lysosomes** originate from the Golgi apparatus.
- They contain digestive enzymes which hydrolyze macromolecules into monomers.
- Food molecules enter the cell by **phagocytosis**—a phagosome is formed.
- Phagosomes fuse with primary lysosomes to form **secondary lysosomes**.
- Enzymes in the secondary lysosome hydrolyze the food molecules.

Lysosomes Isolate Digestive Enzymes from the Cytoplasm



Lysosomes

- Lysosomes also digest cell materials (**autophagy**). 2016 Nobel Prize in Medicine
- Cell components are frequently destroyed and replaced by new ones.



Breakdown of long fatty acids
Found abundantly in liver

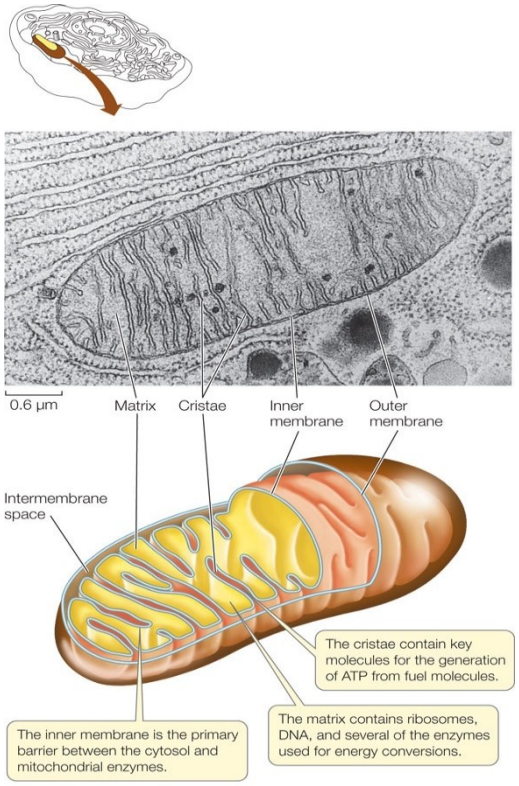
Mitochondria



- In the **mitochondria**, energy in fuel molecules is transformed to the bonds of energy-rich ATP (**cellular respiration**).
- Cells that require a lot of energy have many mitochondria.

Mitochondria

- Mitochondria have two membranes.
- The inner membrane folds inward to form cristae. This creates a large surface area for the proteins involved in cellular respiration reactions.
- The mitochondrial matrix contains enzymes, DNA, and ribosomes.



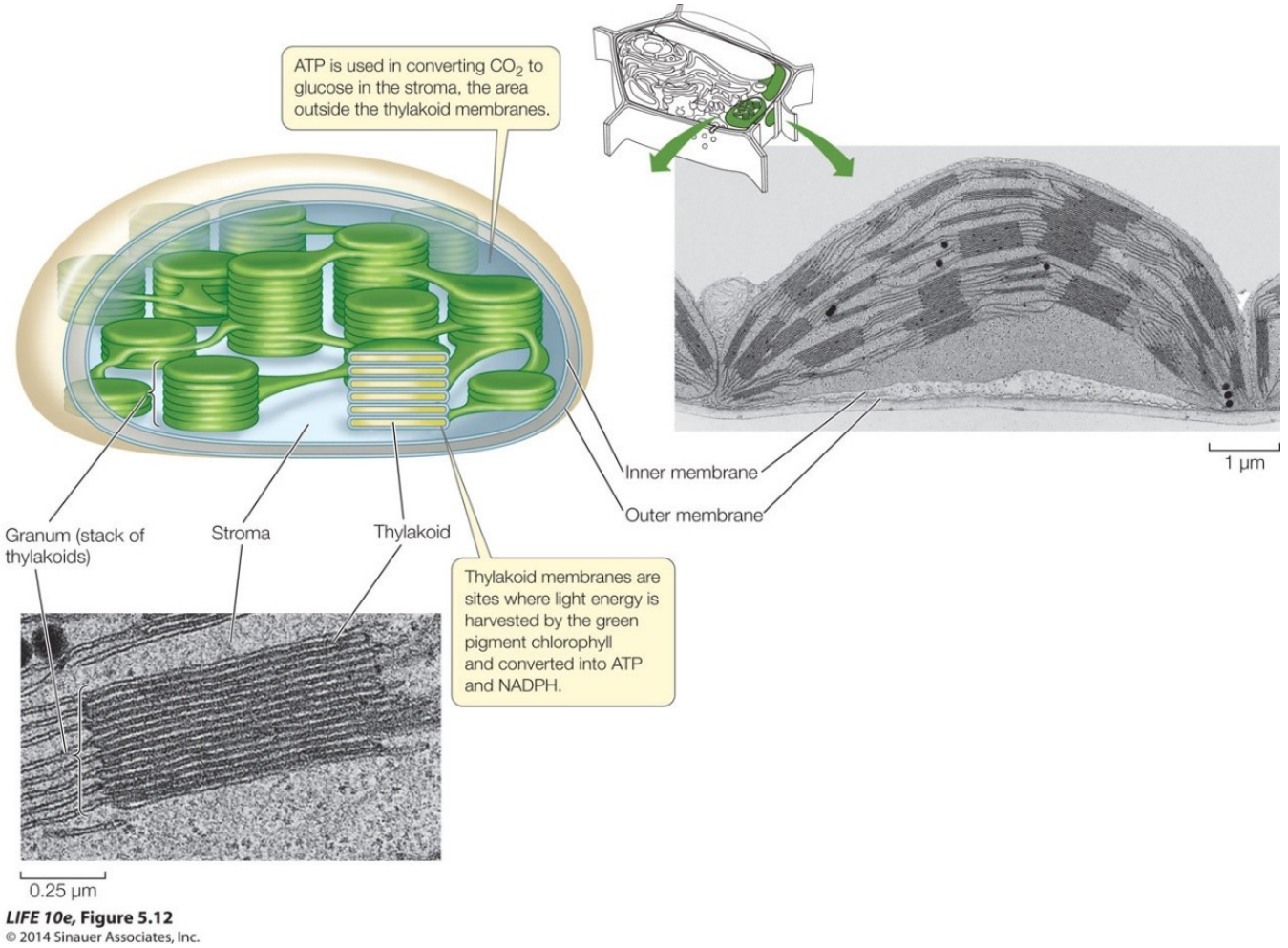
***LIFE 10e*, Figure 5.11**

© 2014 Sinauer Associates, Inc.

Chloroplasts

- Plastids occur only in plants and some protists.
- **Chloroplasts**: Site of photosynthesis—light energy is converted to the energy of chemical bonds.
- Chloroplasts have a double membrane.

Chloroplasts Feed the World

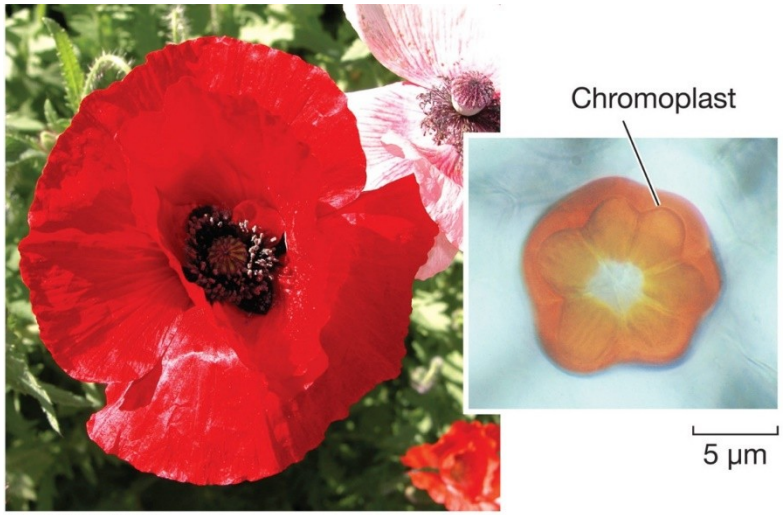


***LIFE 10e*, Figure 5.12**

© 2014 Sinauer Associates, Inc.

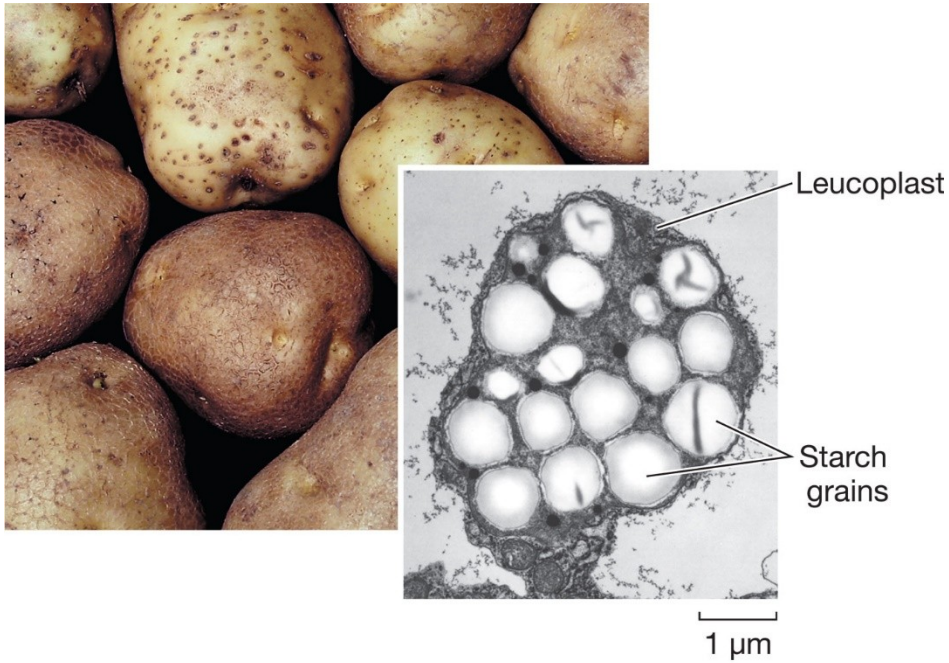
Other plastids

- Other plastids:
- Chromoplasts contain red, orange, and yellow pigments—gives color to flowers.



5.3 What Features Characterize Eukaryotic Cells?

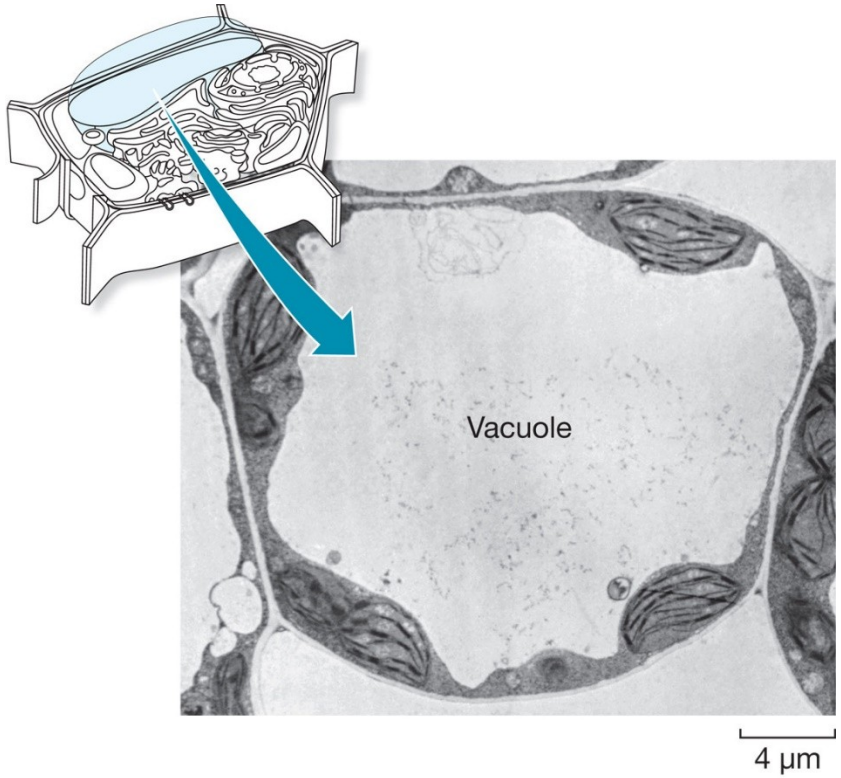
- Leucoplasts store starches and fats.



Vacuoles

- Plant and protist (one celled eukaryotes) cells have **vacuoles**:
- Store waste products and toxic compounds; some may deter herbivores.
- Provide structure for plant cells—water enters the vacuole by osmosis, creating turgor pressure.
- Store anthocyanins (pink and blue pigments) in flowers and fruits; the colors attract pollinators.
- Vacuoles in seeds have digestive

Vacuoles in Plant Cells Are Usually Large to maintain turgor pressure



LIFE 10e, Figure 5.13
© 2014 Sinauer Associates, Inc.

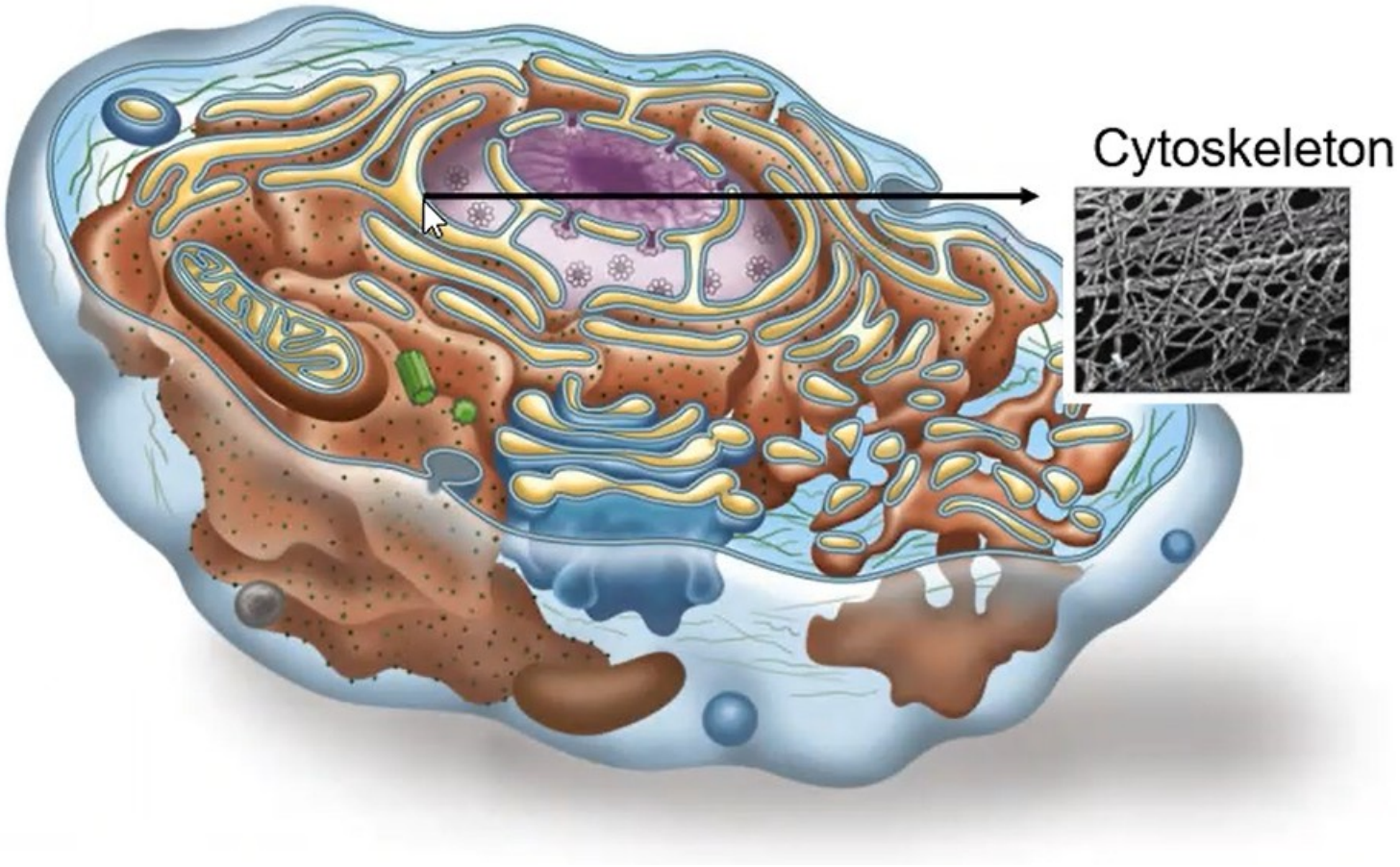
- http://learn.genetics.utah.edu/content/cells/insideacell/

Cytoskeleton

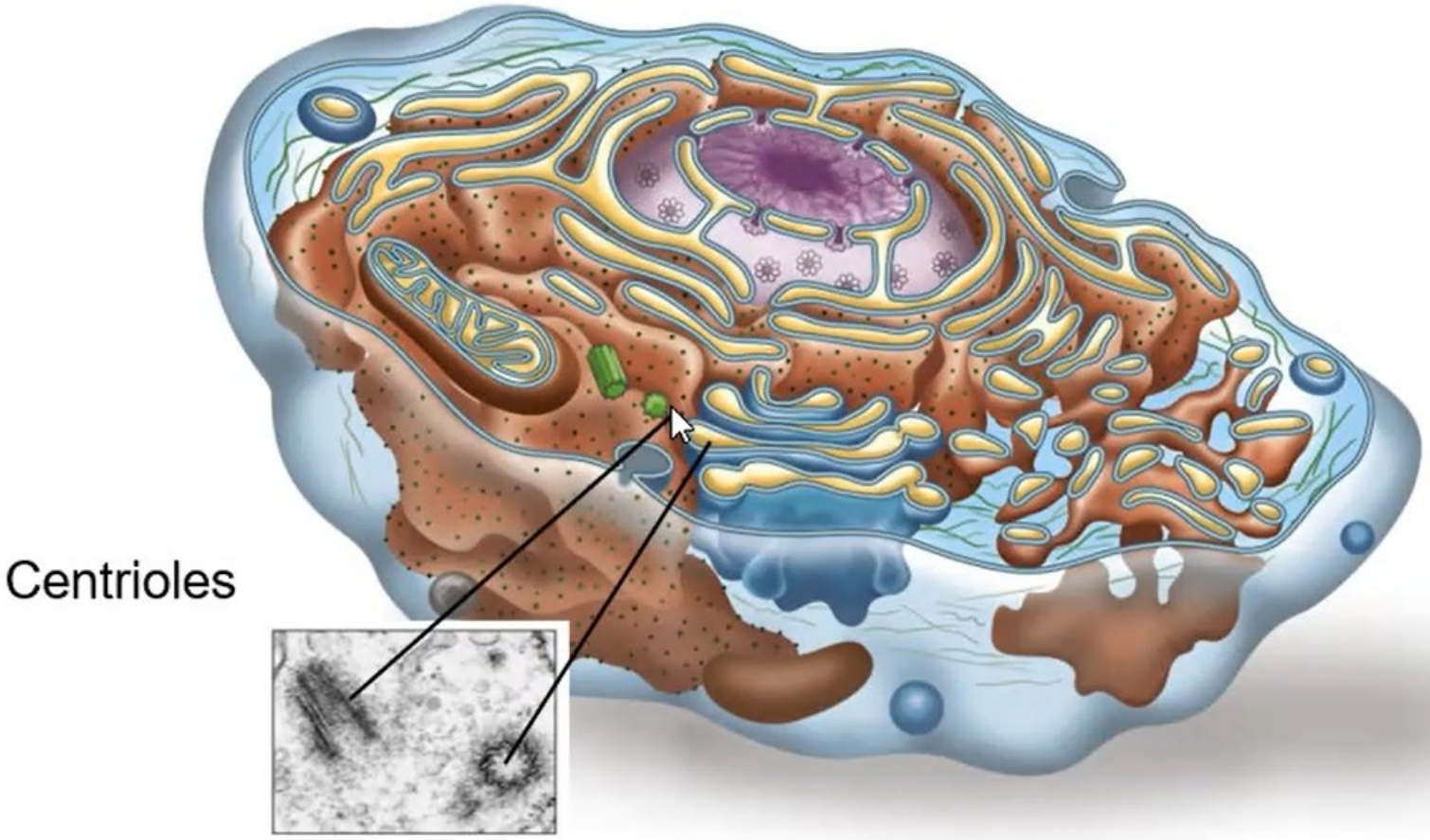
Cytoskeleton

- The highway system in the cell
- Supports and maintains cell shape
- Holds oranelles in position
- Moves organelles in position
- Involved in cytoplasmic streaming
- Interacts with extracellular structures to hold cell in place

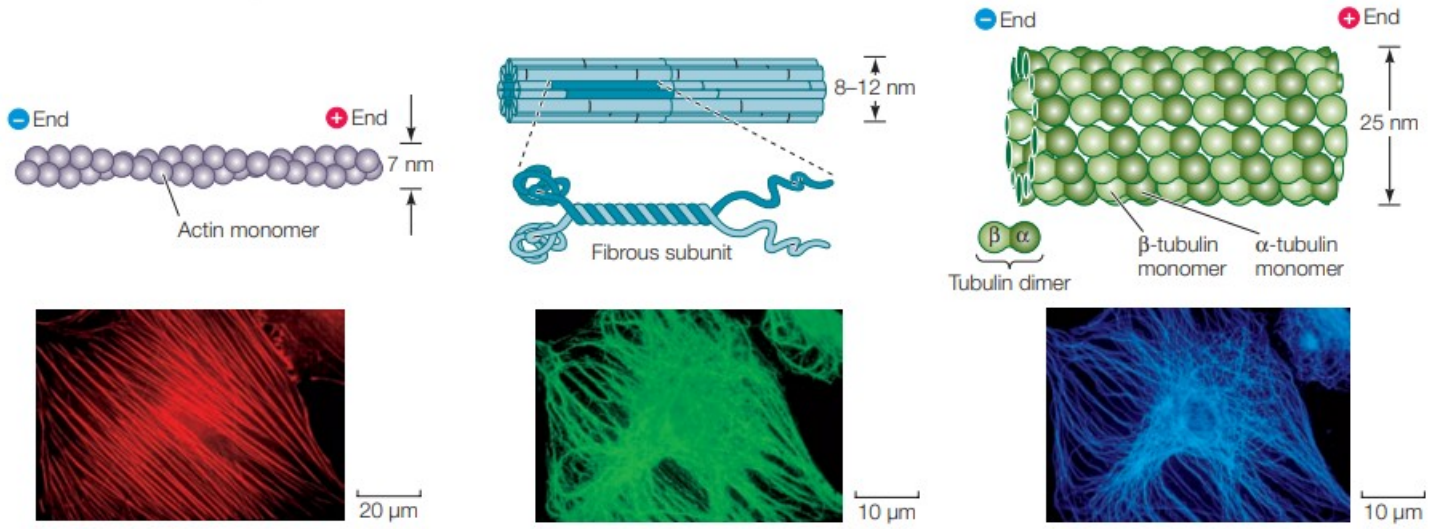
A cytoskeleton composed of microtubules, intermediate filaments, and microfilaments supports the cell and is involved in cell and organelle movement.



Centrioles are associated with nuclear division.



The components of Cytoskeleton



(A) **Microfilaments**
Made up of strands of the protein actin; often interact with strands of other proteins.

(B) **Intermediate filaments**
Made up of fibrous proteins organized into tough, ropelike assemblages that stabilize a cell's structure and help maintain its shape.

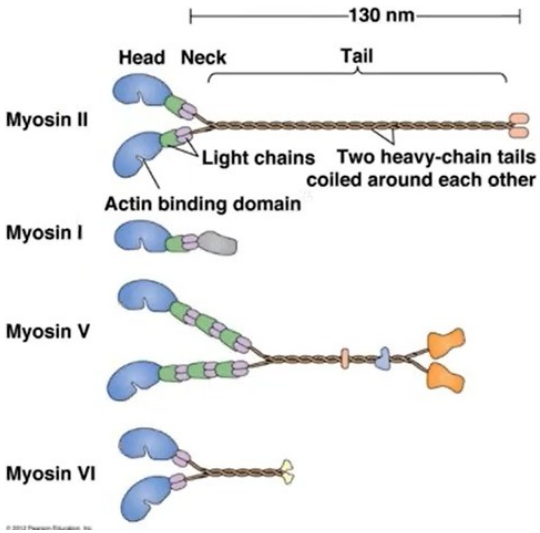
(C) **Microtubules**
Long, hollow cylinders made up of many molecules of the protein tubulin. Tubulin consists of two subunits, α-tubulin and β-tubulin.

5.17 The Cytoskeleton Three highly visible and important structural components of the cytoskeleton are shown here in detail. These structures maintain and reinforce cell shape and contribute to cell movement.

1. Microfilaments

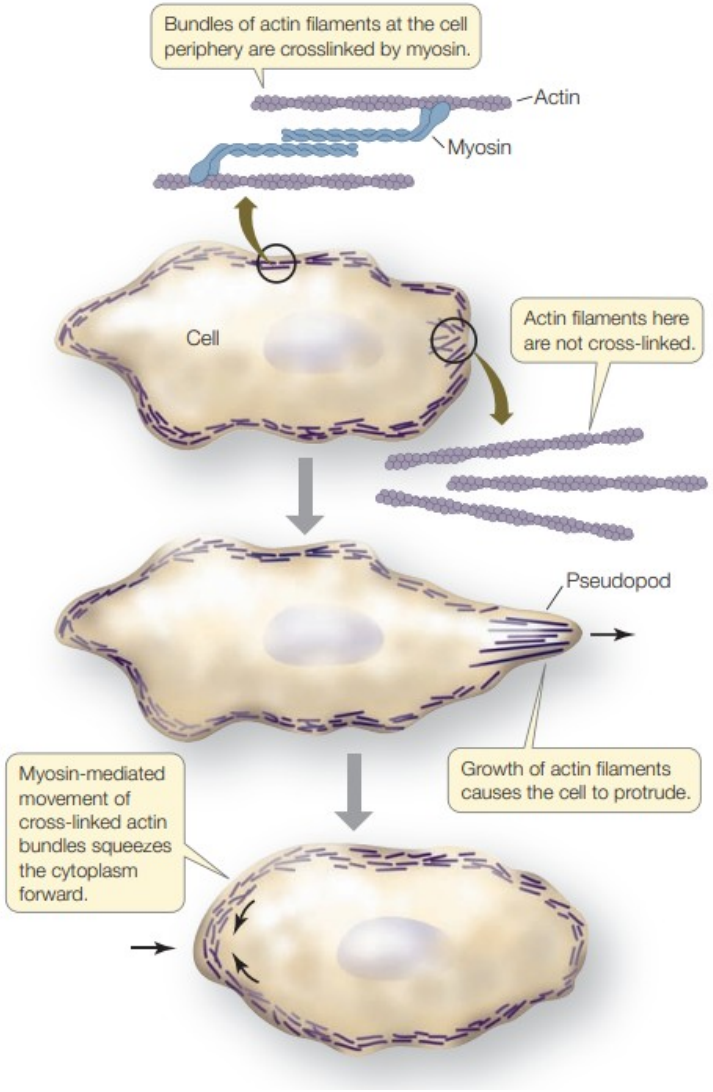
Actin-Based Cell Movement and the Myosins

- Movements of molecules and the other cell components also occur along another system in the cell- actin cytoskeleton
- Functions iclude muscle contraction, cell movement, phagocytosis, vesicle transport



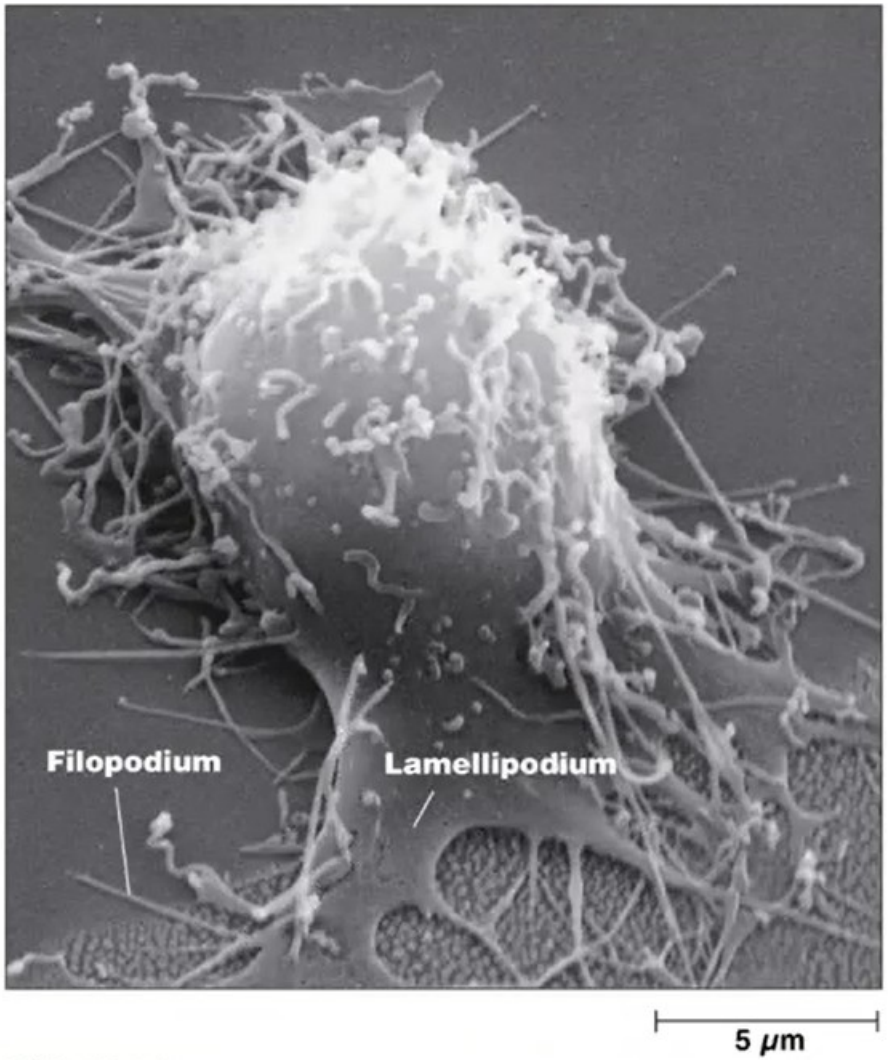
The globular head binds actin and uses the energy of ATP hydrolysis to move along the filament

Microfilaments and Cell Movements



- Microfilaments are required for the movement of most cells in animals
- Cell crawling involves distinct events: extension of a protrusion, attachment to substrate, and generation of the tension, which pulls the cell forward

Microfilaments and Cell Movements



© 2012 Pearson Education, Inc.

- A thin sheet of cytoplasm is a l**amellipodium** and a thin-pointed protrusion is a **filopodium**

Cell attachment

- Attachment or adhesion of the cell to the substrate is necessary for cell movement
- New sites of attachment are formed at the front of the cell, and the contacts at the rear must be broken
- Attachment sites are complex structures involving a number of proteins

Chemotaxis is a Directional Movement in Response to a Graded Stimulus

- *Directional migration* occurs through the formation of protrusions predominantly on one side of a cell

- Diffusible molecules can act as cues for directional migration; when a cell moves in reponse to a chemical gradient, it is called **chemotaxis.**

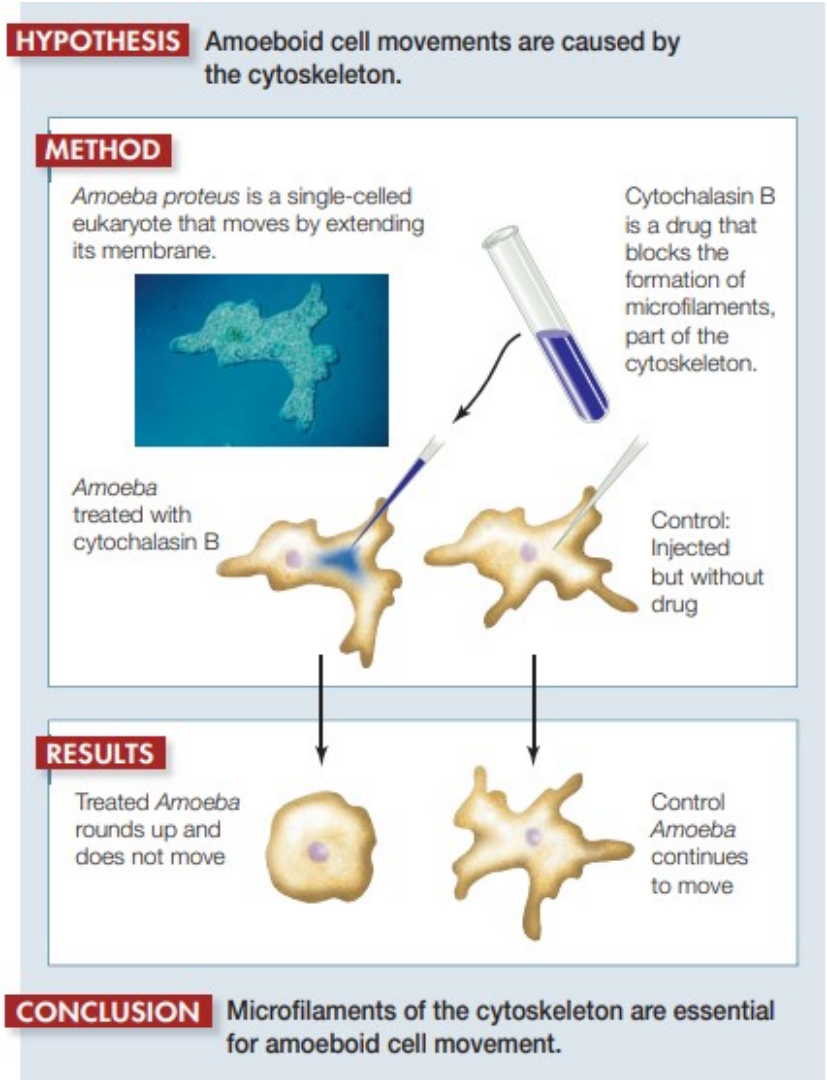
Chemotaxis

- *Chemoattractants: cells move toward a higher concentration of a diffusible molecules*
- *Chemorepellants: cells move toward a lower concentration of a diffusible molecules*
- *Binding of the molecules to cell surface receptors (G-protein linked receptors) leads to corresponding cytoskeletal changes*
- https://www.youtube.com/watch?v=Z_mXDvZQ6dU
- https://www.youtube.com/watch?v=LXno0vcW2Go
- https://www.youtube.com/watch?v=DggSgZO6iZQ

The Role of Microfilaments in Cell Movement- Showing Cause and Effect in Biology

5.23 The Role of Microfilaments in Cell Movement—Showing Cause and Effect in Biology

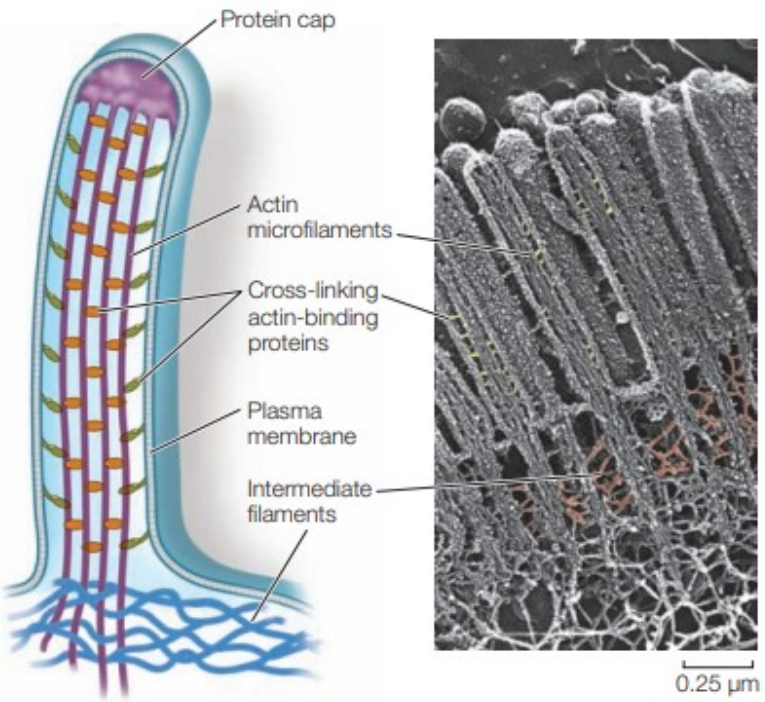
After a test tube demonstration that the drug cytochalasin B prevented microfilament formation from monomeric precursors, the question was asked: Will the drug work like this in living cells and inhibit cell movement in *Amoeba*? Complementary experiments showed that the drug did not poison other cellular processes.



Microvilli are made up of microfilaments

- In some cells, microfilaments form a meshwork just inside the plasma membrane.

- This provides a structure, for example in the microvilli that line the human intestine.



5.19 Microfilaments for Support Cells that line the intestine are folded into tiny projections called microvilli, which are supported by microfilaments. The microfilaments interact with intermediate filaments at the base of each microvillus. The microvilli increase the surface area of the cells, facilitating their absorption of small molecules.

2. Microtubules

Microtubules

- **Cilia** and eukaryotic **flagella** are made of microtubules in ‘9+2’ array

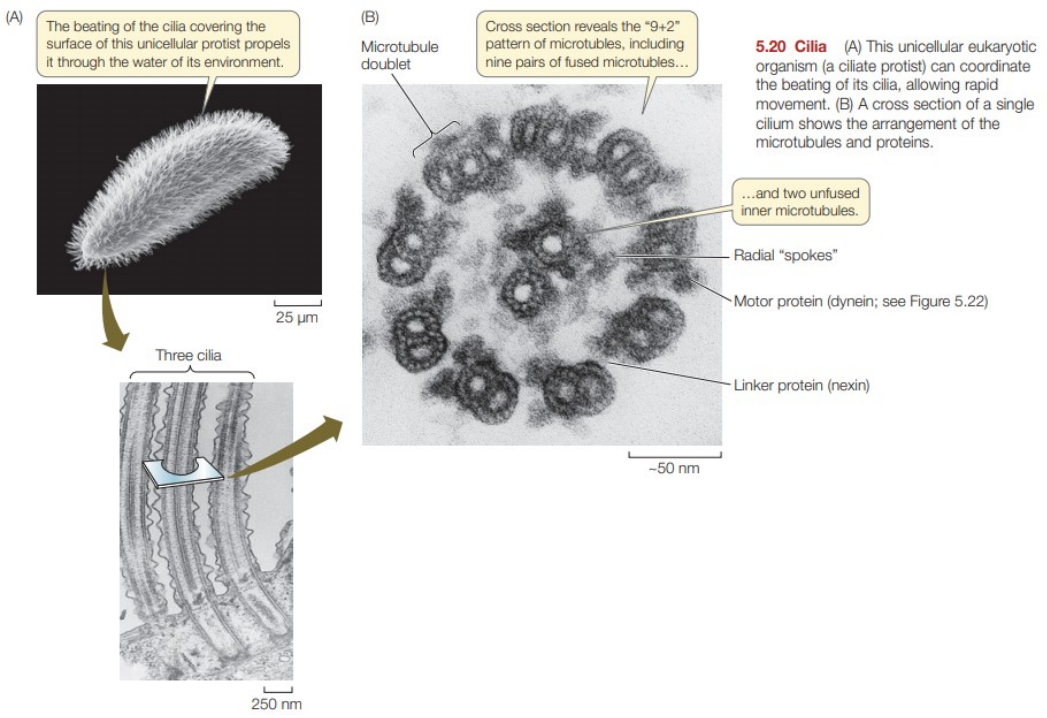
- Cilia- short, usually many present, move with stiff power stroke and flexible recovery stroke

- Flagella- long, usually one or two present, movement is snake-like.

Microtubules

- The motion of **cilia** and **flagella** results from the sliding of the microtubule doublets past one another.

- Dynein binds to microtubule doublets and allows them to slide past each other.

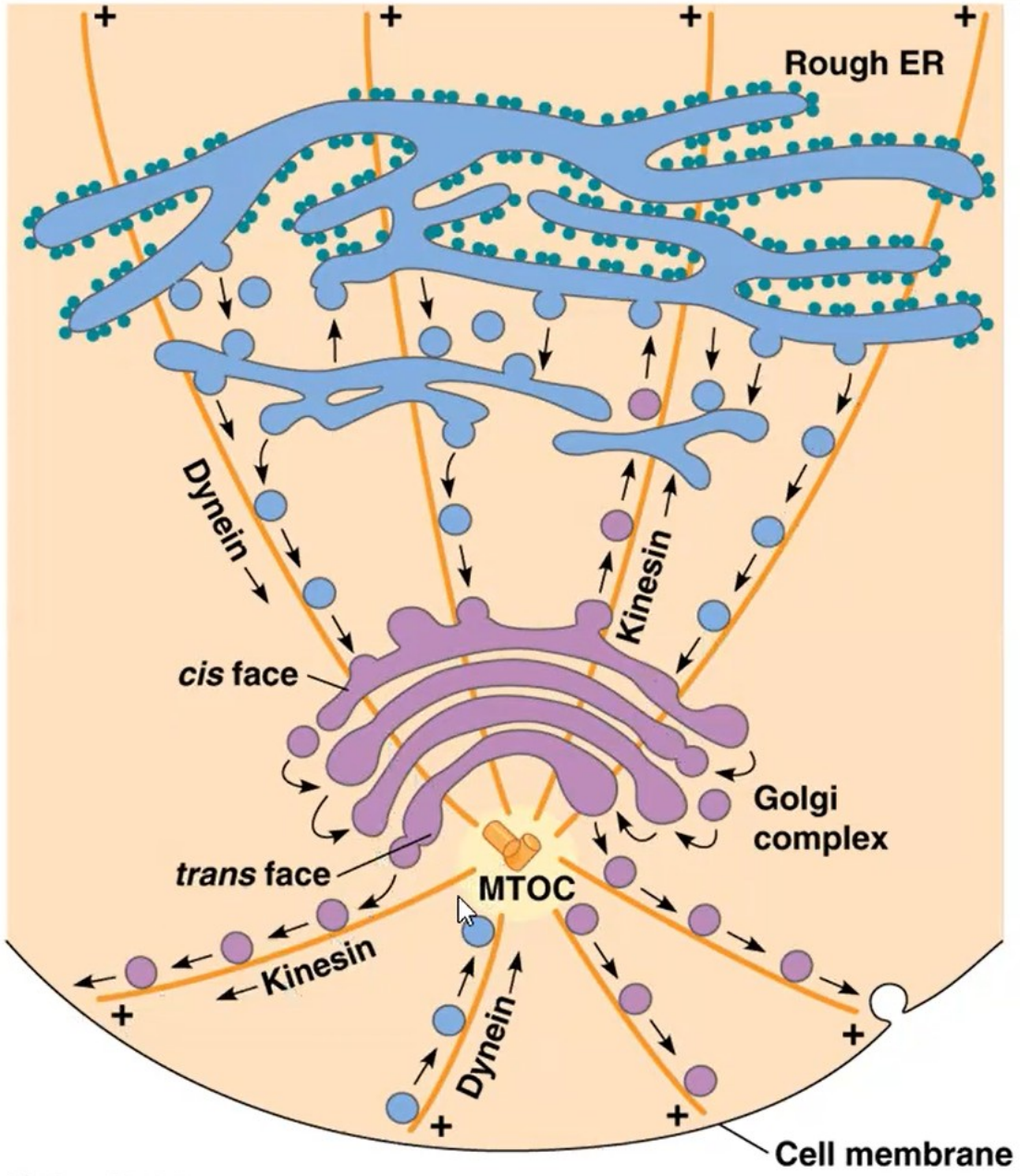


Motor proteins move cargo on microtubules inside the cell

- The motor protein kinesin moves vesicles or organelles from one part of a cell to another

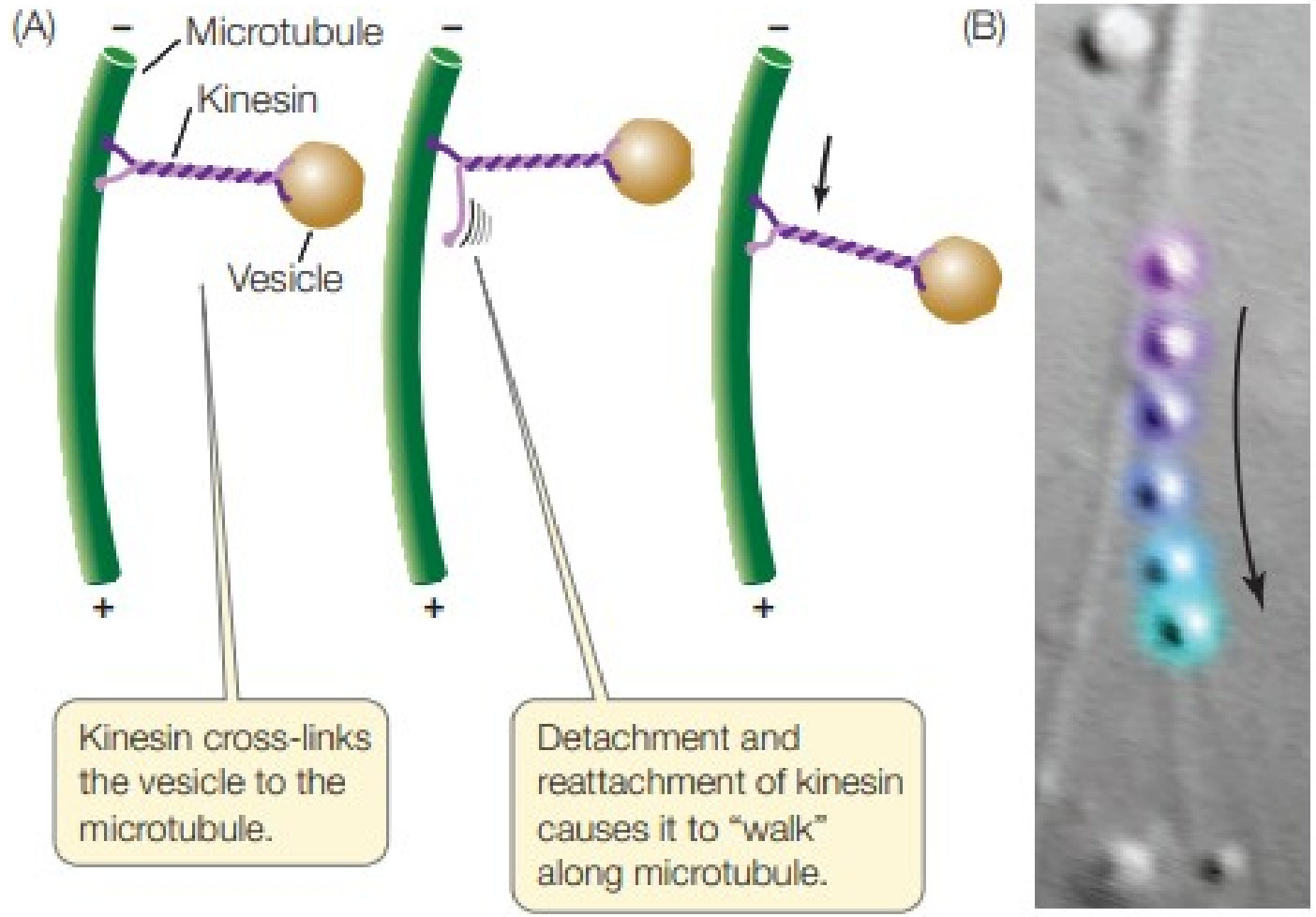
- It binds to a vesicle and «walks» it along by changing shape

https://www.youtube.com/watch?v=plvQCOE9s_k



© 2012 Pearson Education, Inc.

A motor protein pulls vesicles along microtubules



3. Intermediate Filaments

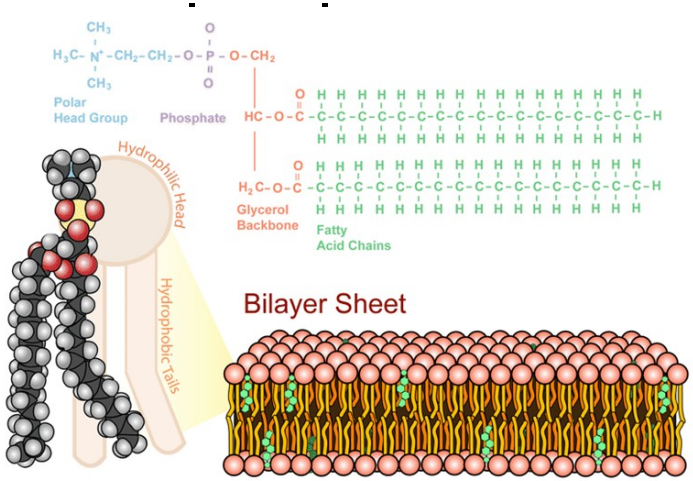
Keratins, Nuclear lamins

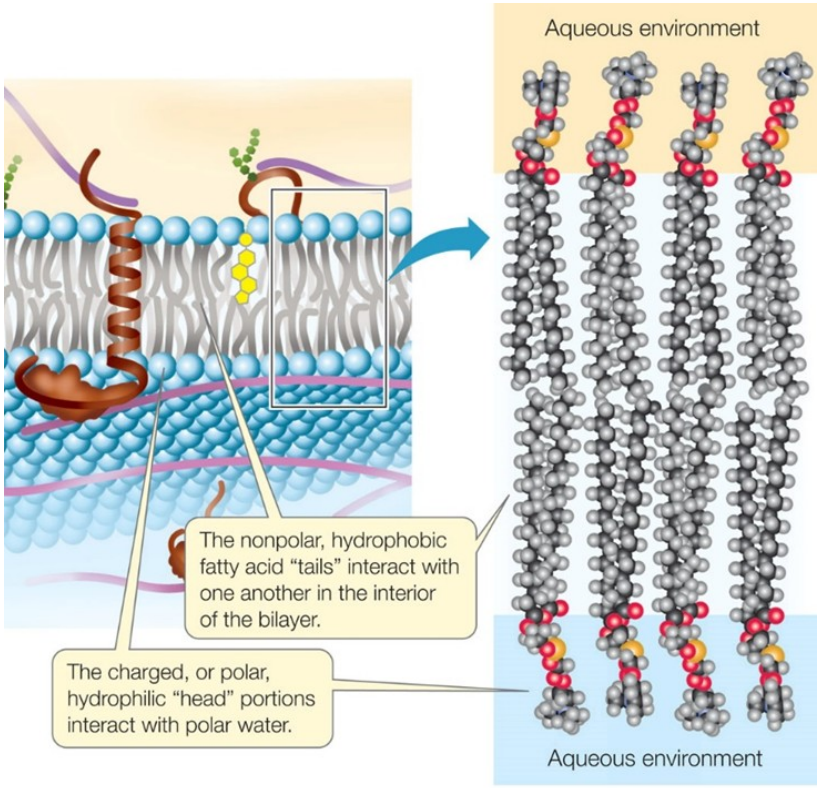
Specialized for different cell types

Cell Membrane

Cell membrane

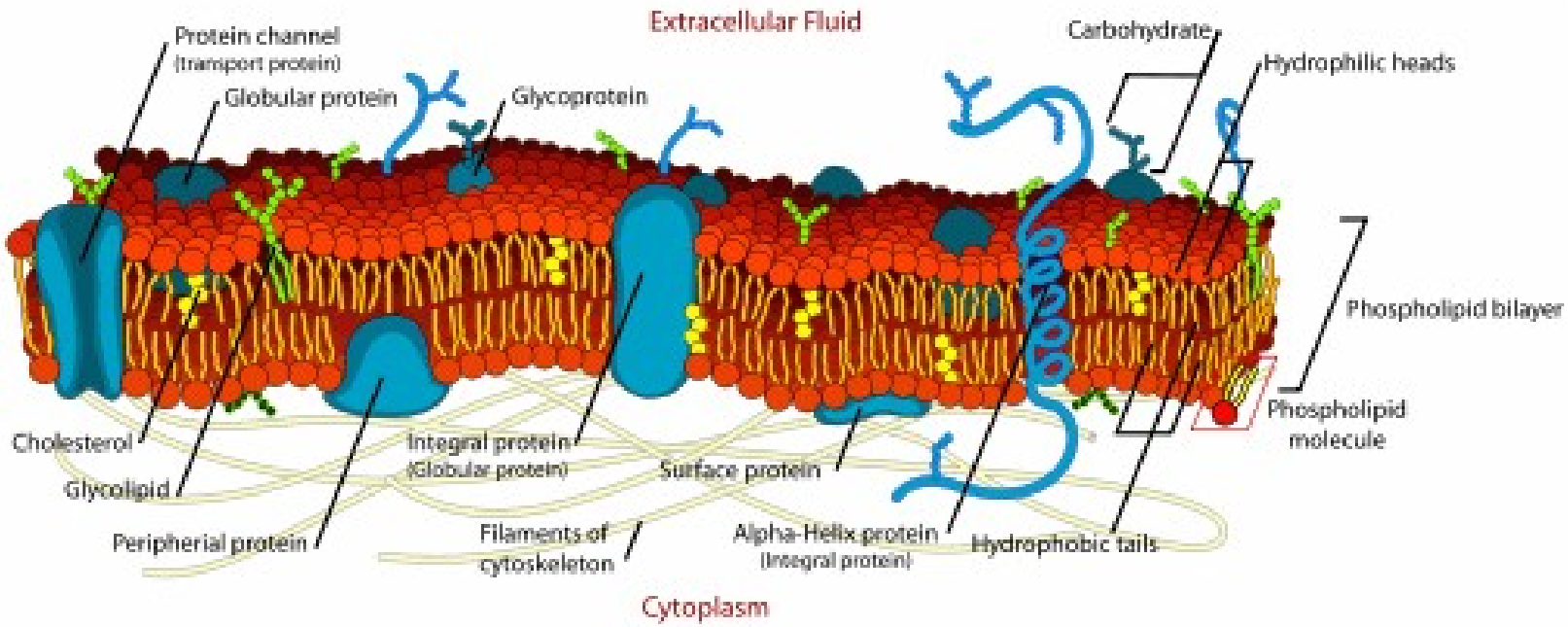
- The **plasma membrane** is the outer surface of every cell and has more or less the same structure in all cells.
- It is made of a **phospholipid bilayer** with embedded proteins and oth





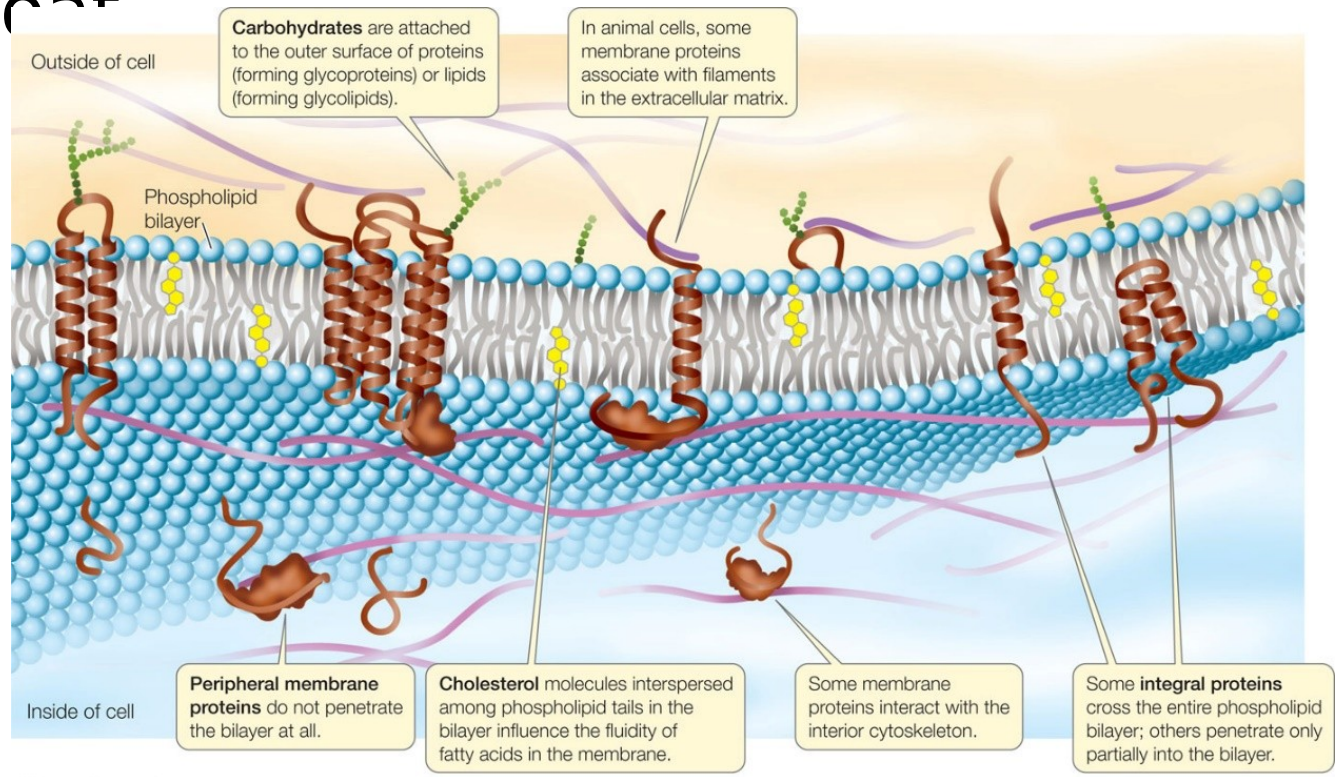
LIFE 10e, Figure 6.2
© 2014 Sinauer Associates, Inc.

Cell membrane



- The plasma membrane:
  - is a selectively permeable barrier
  - allows cells to maintain a constant internal environment
  - is important in communication and receiving signals
  - often has proteins for binding and adhering

- The general structure of biological membranes is known as the **fluid mosaic model**.
- Phospholipids form a bilayer, which is like a "lake" in which a variety of proteins "float."



LIFE 10e, Figure 6.1
© 2014 Sinauer Associates, Inc.

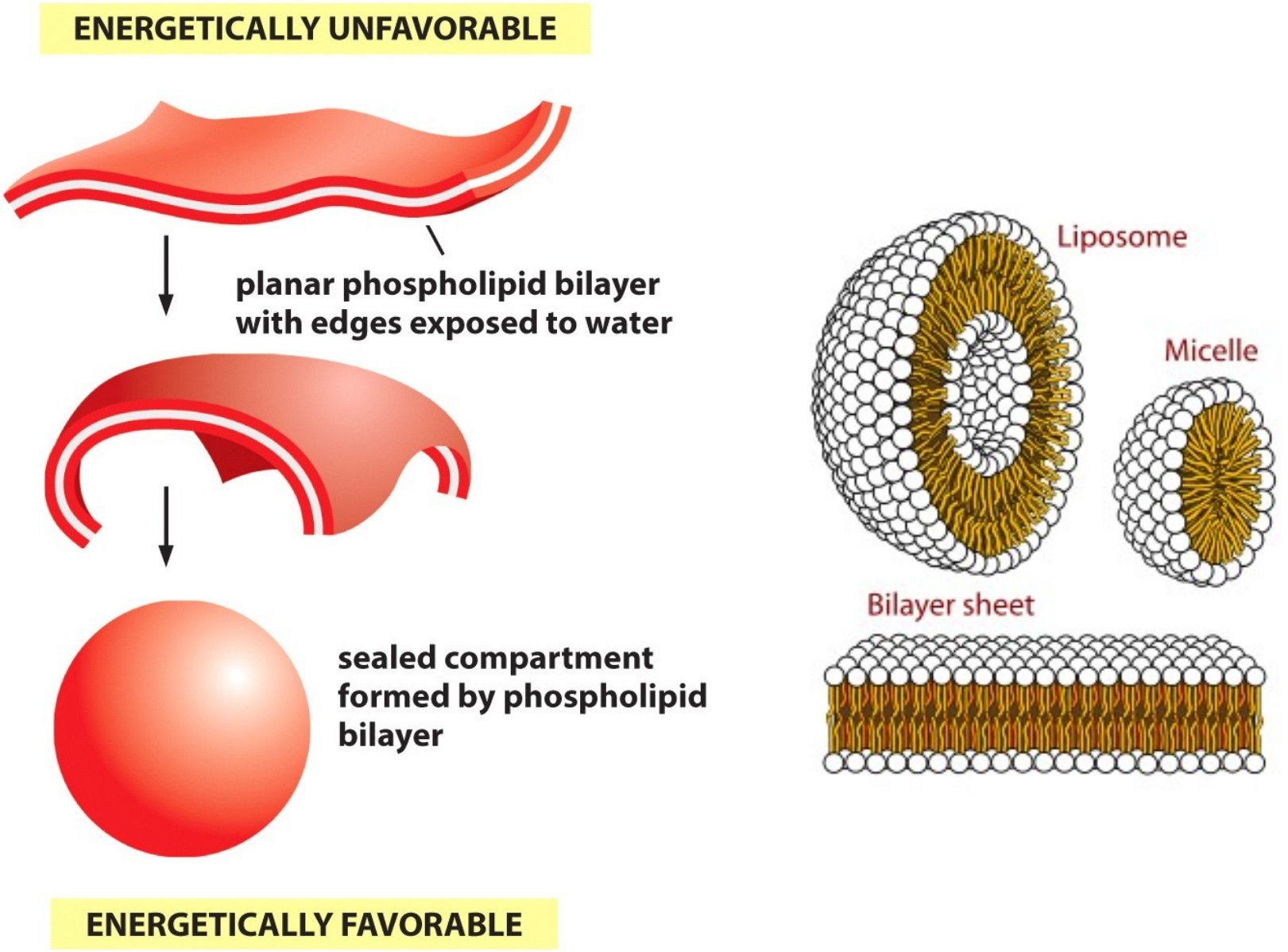
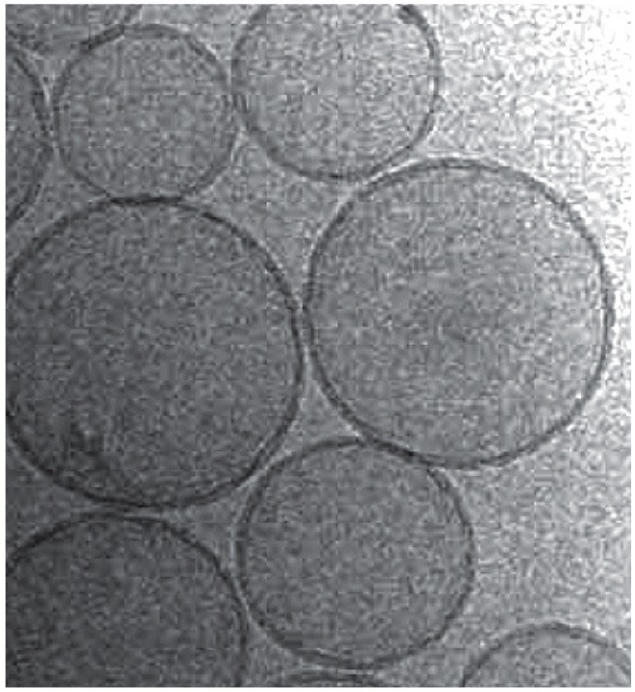
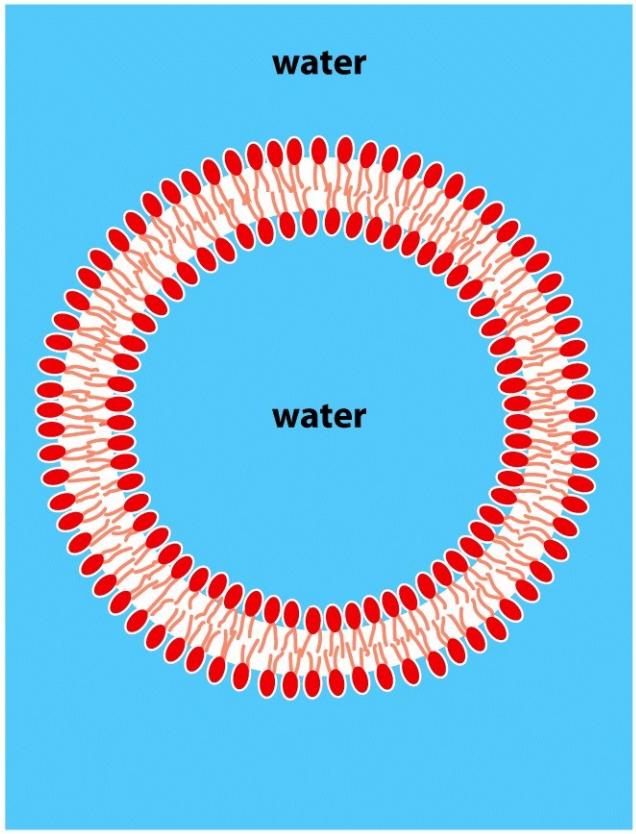


Figure 11-12 Essential Cell Biology, 4th ed. (© Garland Science 2014)





50 nm

25 nm

(A)

(B)

Figure 11-13 Essential Cell Biology, 4th ed. (© Garland Science 2014)

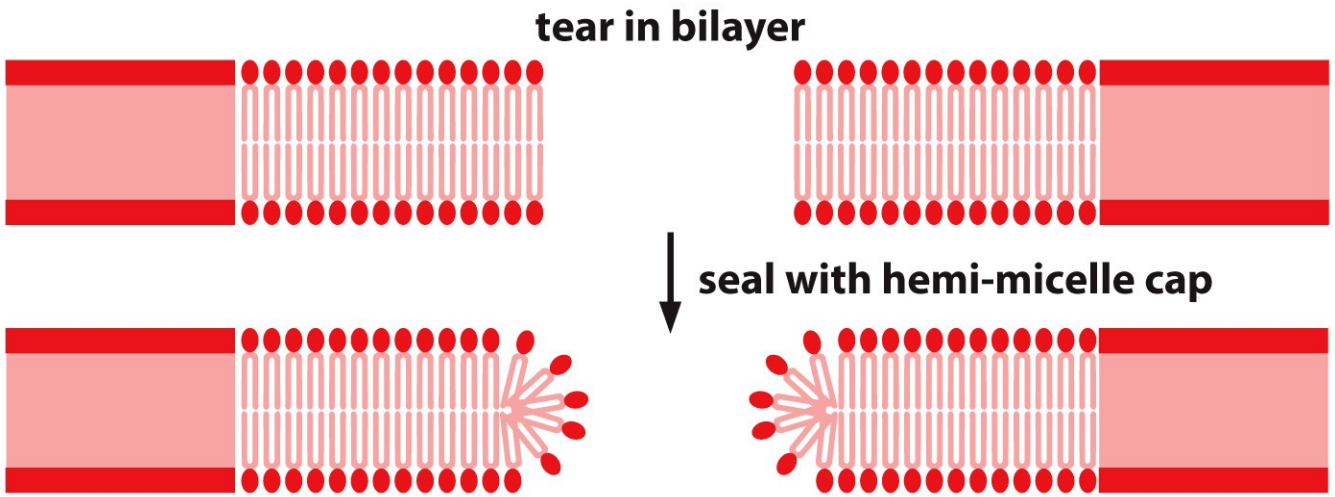
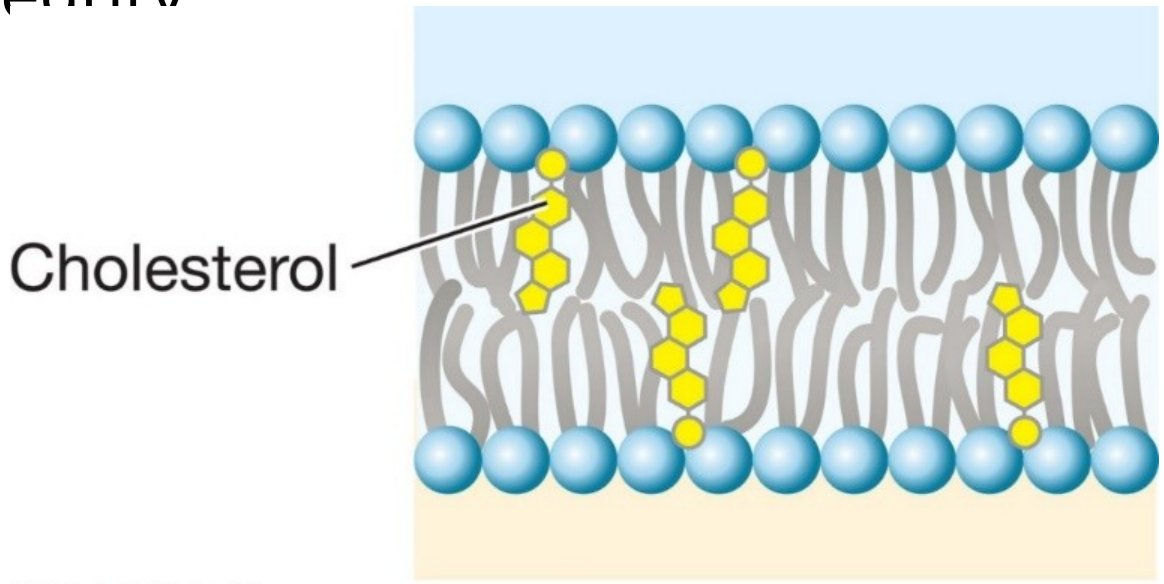


Figure Q10-1 Molecular Biology of the Cell 6e (© Garland Science 2015)

What Is the Structure of a Biological Membrane?

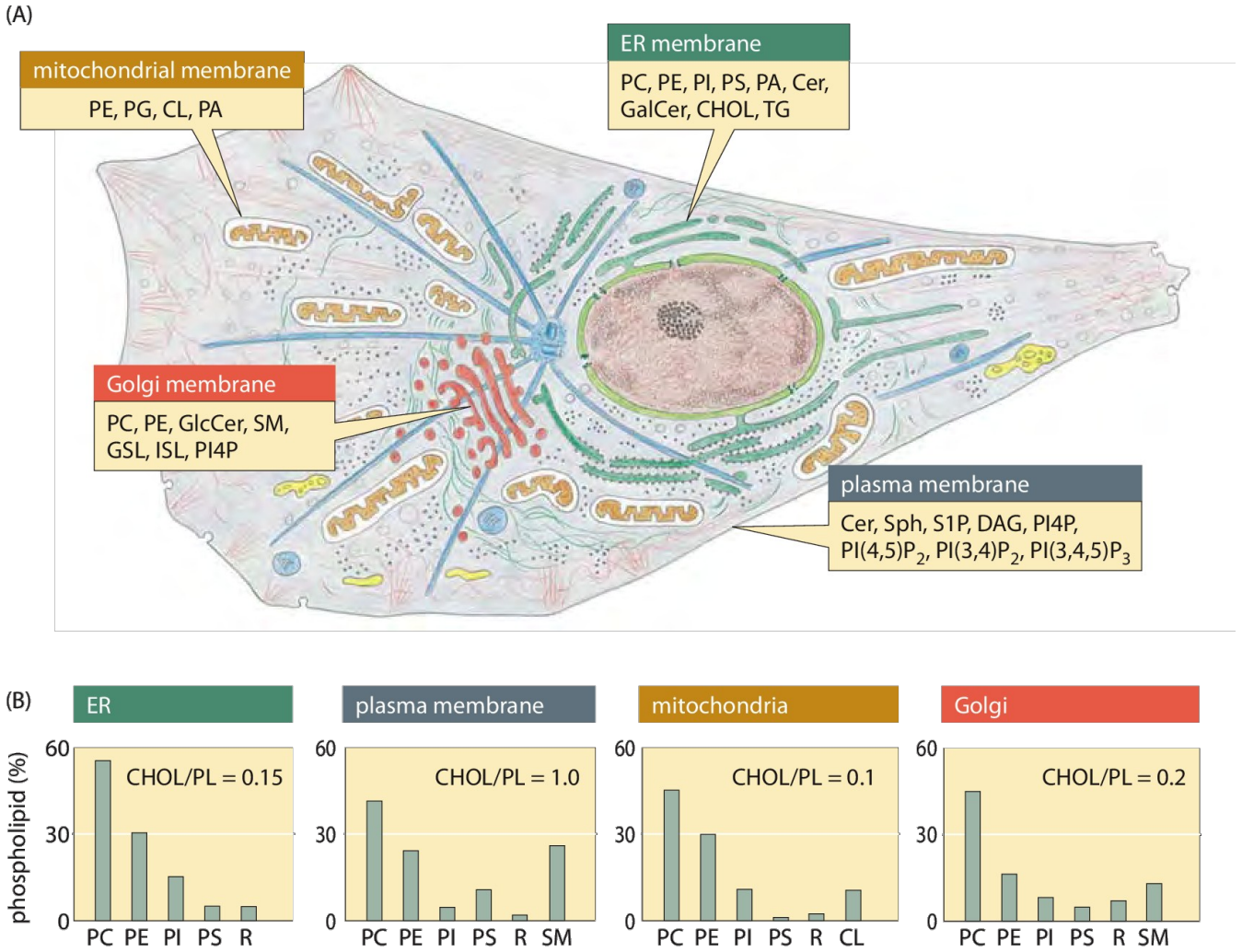
- Lipid composition of membranes vary.
- Phospholipids vary in fatty acid chain length, degree of saturation, and phosphate groups.
- Animal cell membranes may be up to 25% cholesterol, which is important for membrane integrity



What Is the Structure of a Biological Membrane?

- The fatty acid tails make the interior somewhat fluid, allowing lateral movement of molecules.
- Fluidity depends on temperature and lipid composition.
- Long-chain, saturated fatty acids pack tightly, making a less-fluid membrane.
- As temperature decreases, movement of molecules and cellular processes slow. Some organisms change the lipid content of the cell membranes when they get cold.

The phospholipid composition of biological membranes is variable



Phospholipids move around in the membrane

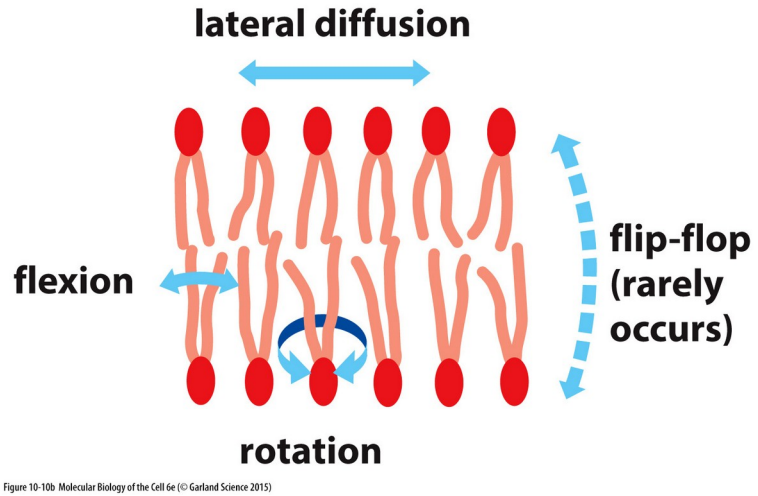


Figure 10-10b Molecular Biology of the Cell 6e (© Garland Science 2015)

This needs specialized proteins to move phospholipids from one layer to the other

Membrane Proteins

- Membranes also contain proteins; the number varies depending on membrane function. Two categories:

1. Peripheral membrane proteins lack exposed hydrophobic groups and do not penetrate the bilayer.
2. Integral membrane proteins have hydrophobic and hydrophilic regions or domains. Some extend across the lipid bilayer; others are partially embedded.

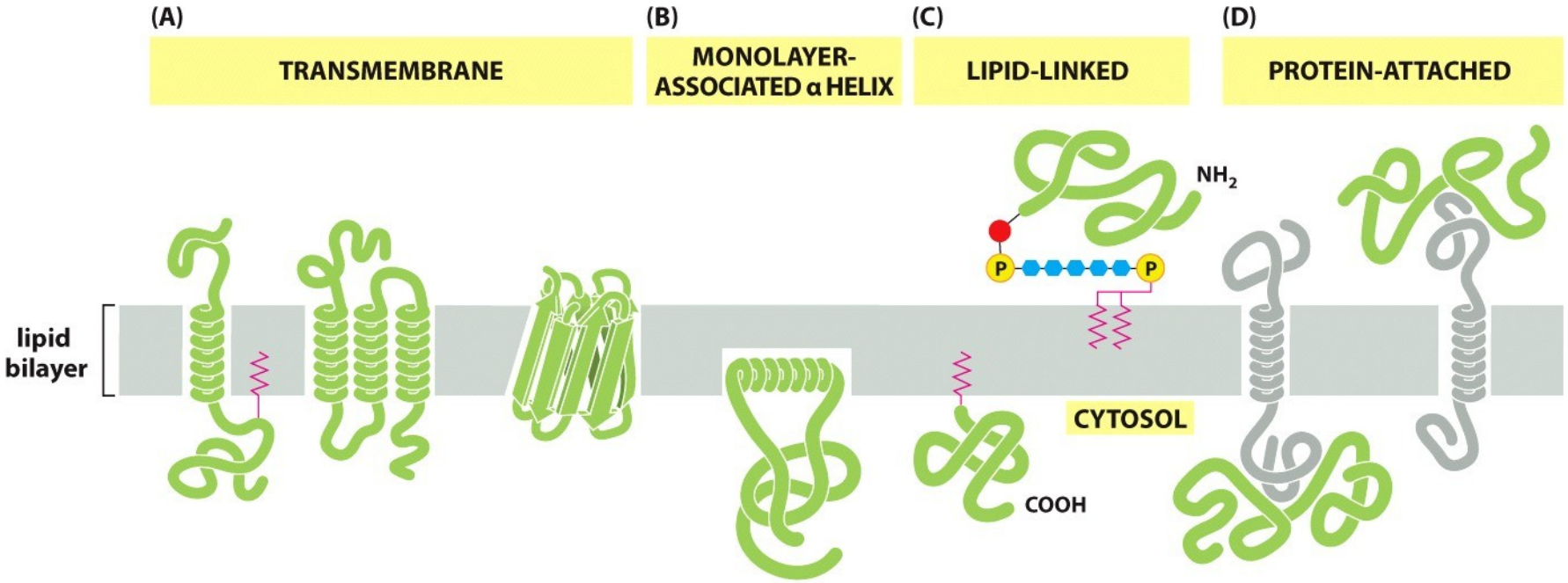
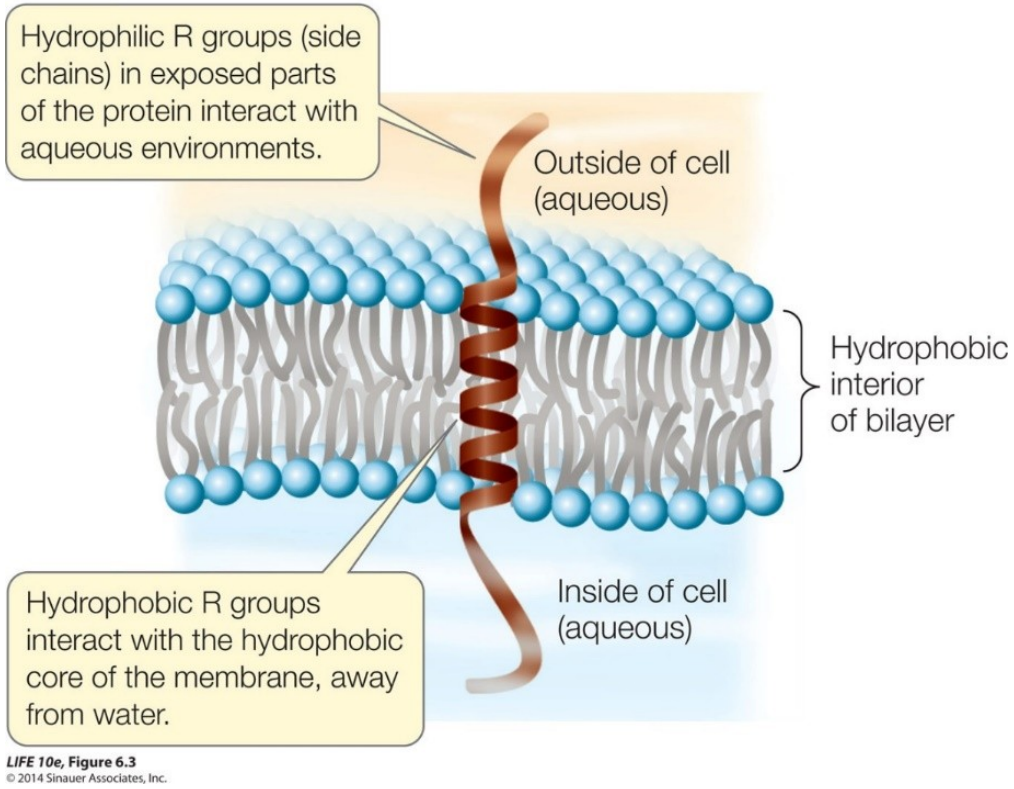


Figure 11-20 Essential Cell Biology, 4th ed. (© Garland Science 2014)

Integral Membrane Proteins



LIFE 10e, Figure 6.3
© 2014 Sinauer Associates, Inc.

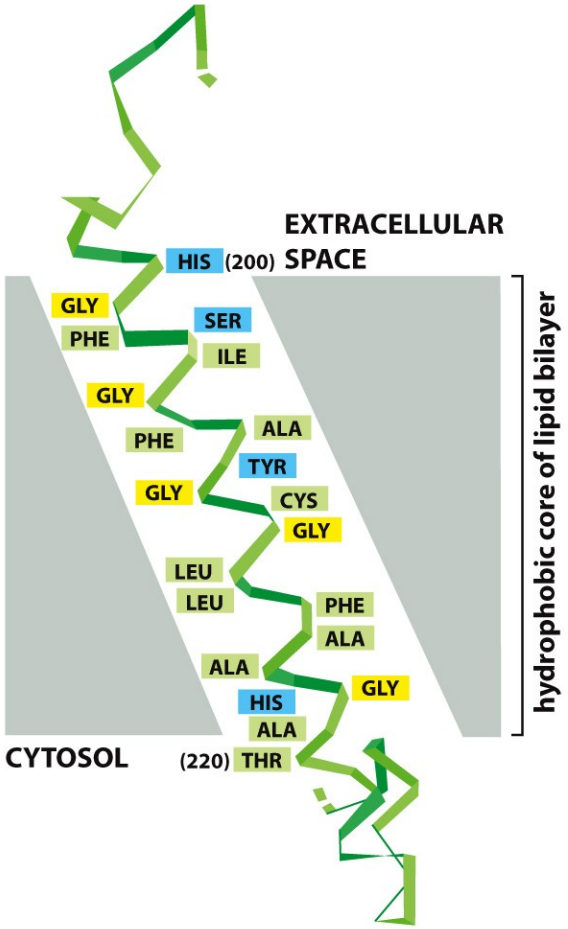


Figure 10-19 Molecular Biology of the Cell 6e (© Garland Science 2015)

Amino acid composition of a membrane protein can be used to predict how many passes that protein makes through the membrane

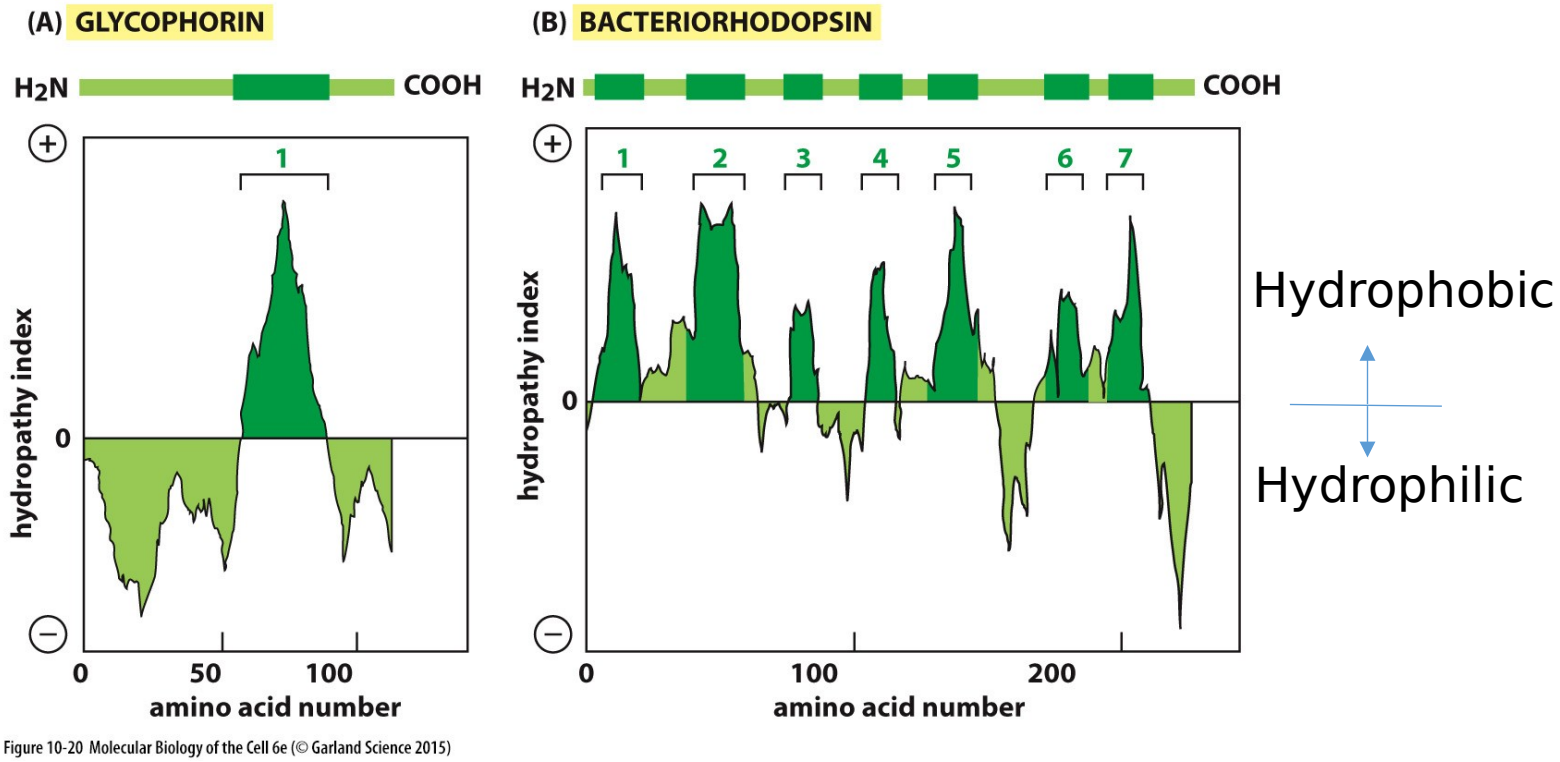


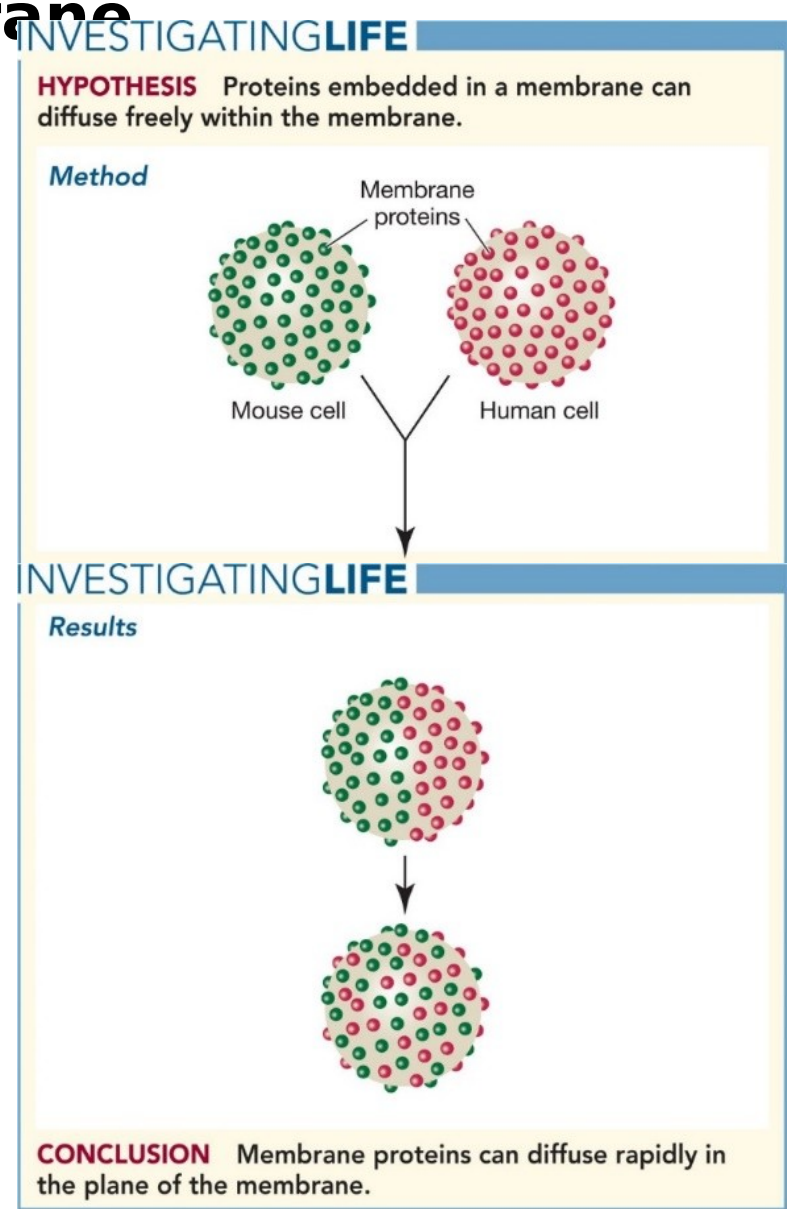
Figure 10-20 Molecular Biology of the Cell 6e (© Garland Science 2015)

Membrane proteins and lipids interact with each other

- The proteins and lipids interact noncovalently.
- But some membrane proteins have lipid groups covalently attached and are tethered to the lipid bilayer.
- Some membrane proteins can move freely within the bilayer, while some are anchored to a specific region.

- How do you test if membrane proteins are mobile?

When cells are fused experimentally, some proteins from each cell distribute themselves uniformly around the membrane



Membrane proteins are mobile

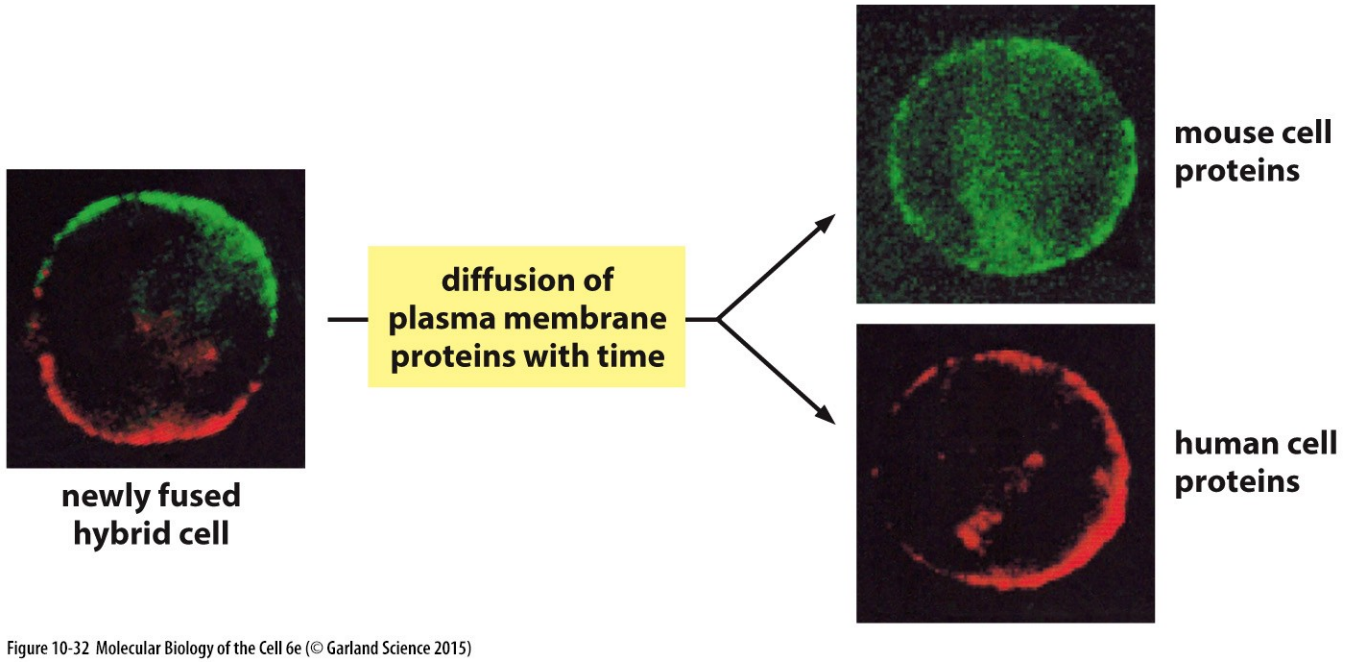


Figure 10-32 Molecular Biology of the Cell 6e (© Garland Science 2015)

Membrane proteins are mobile

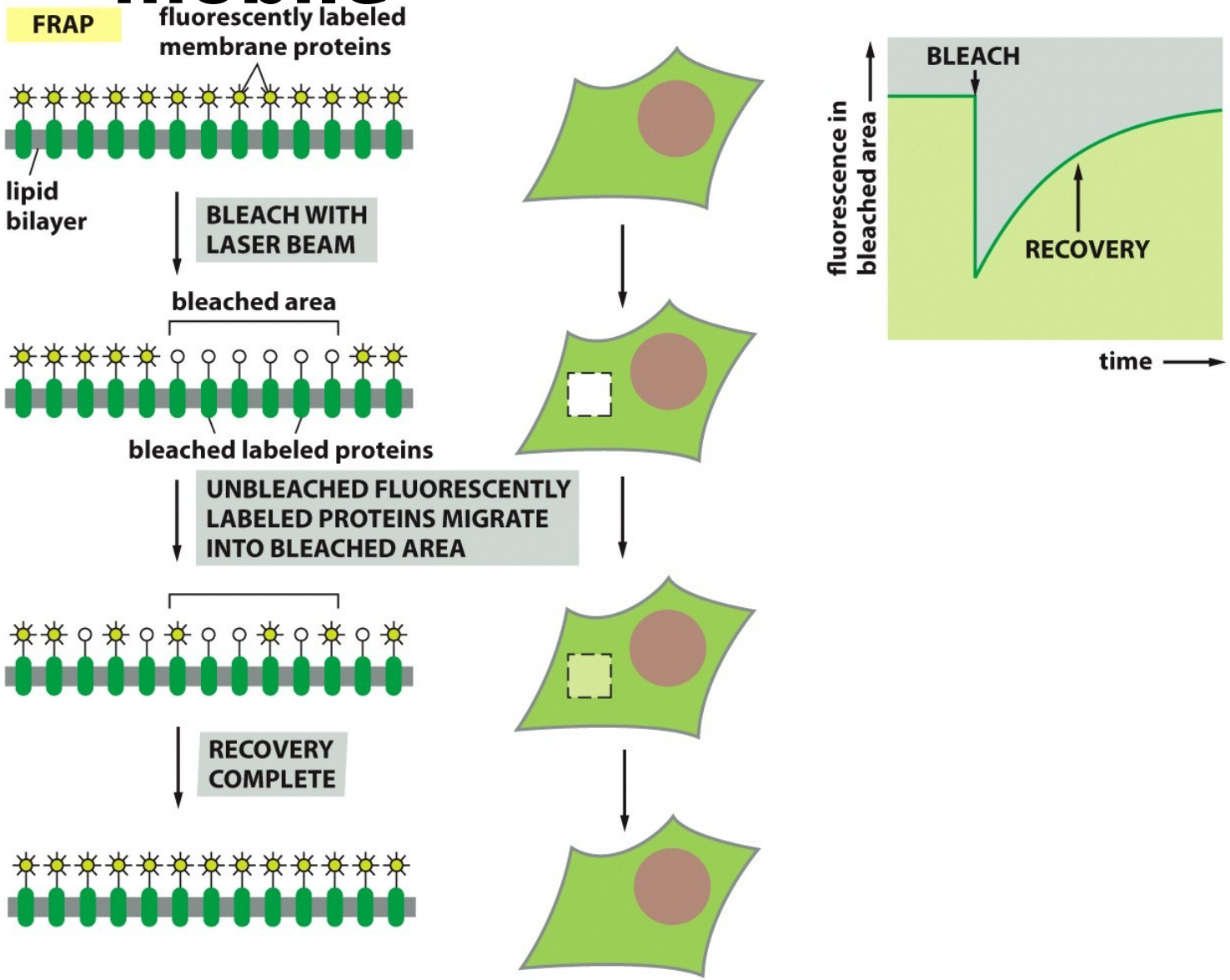
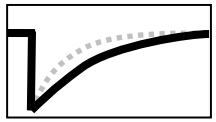
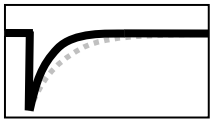


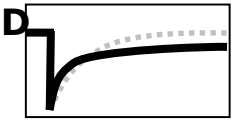
Figure 11-34 Essential Cell Biology, 4th ed. (© Garland Science 2014)

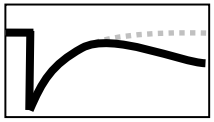
Under a different condition (e.g. when a signaling pathway is turned on), the protein binds to a few other proteins and protein complexes in the plasma membrane. As a result, its lateral diffusion coefficient is reduced and a significant fraction of the molecules of this protein are immobilized in large aggregates. How would you expect the FRAP curve to change under these new conditions? The dotted line in each graph is the original curve shown for comparison.

A 

B 

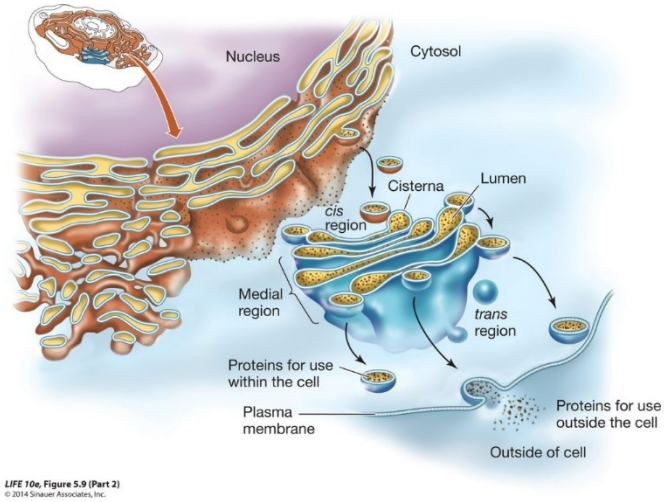
C 

D 

E 

Membrane dynamics

- Membranes are dynamic and are constantly forming, transforming, fusing, and breaking down.



LIFE 10e, Figure 5.9 (Part 2)
© 2014 Sinauer Associates, Inc.

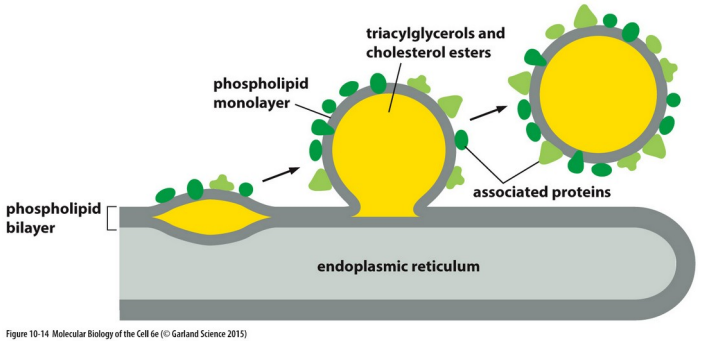


Figure 10-14 Molecular Biology of the Cell 6e (© Garland Science 2015)

ow do proteins that get secreted out get to plasma membran
ow do membrane proteins get to plasma membrane?

Membrane proteins and lipids are usually modified

- Membranes also have carbohydrates on the outer surface that serve as recognition sites for other cells and molecules.
- **Glycolipids**—carbohydrate + lipid
- **Glycoproteins**—carbohydrate + protein

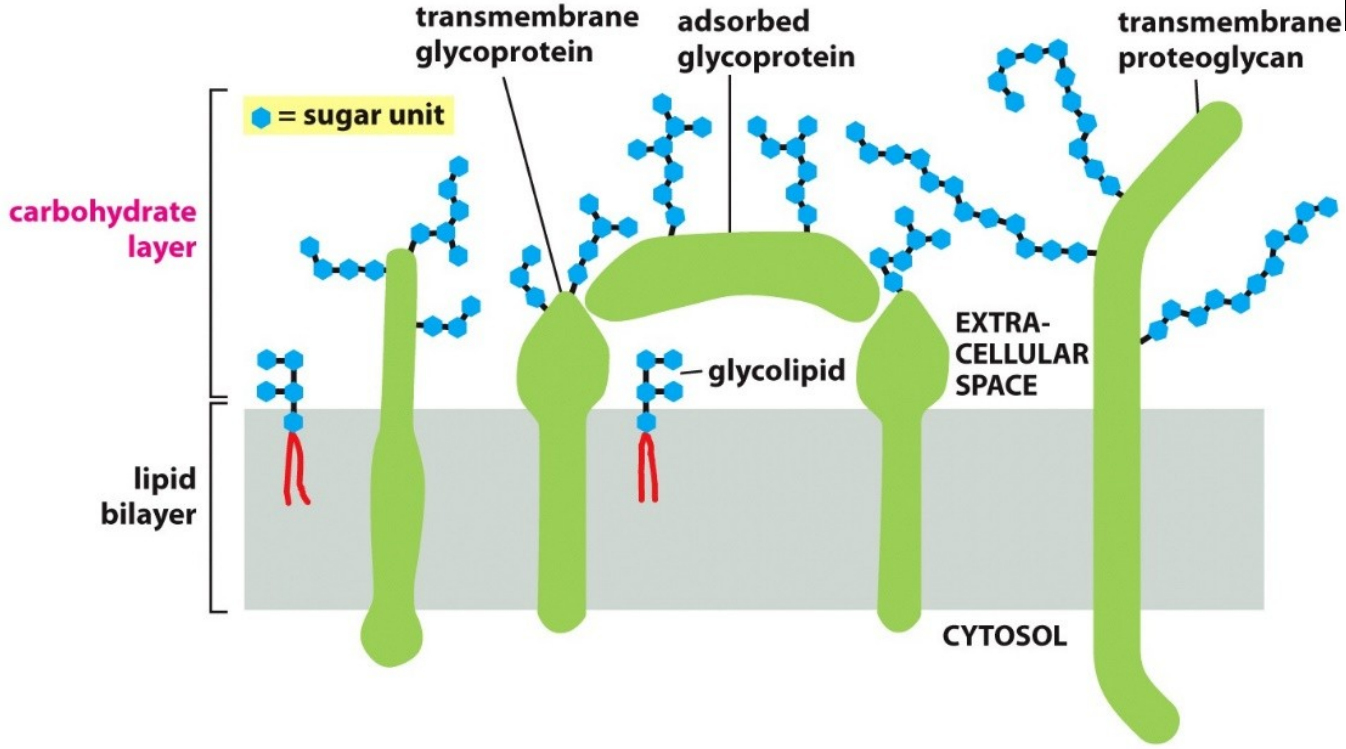


Figure 11-33 Essential Cell Biology, 4th ed. (© Garland Science 2014)

- http://www.xvivo.net/animation/the-inner-life-of-the-cell/